CS177 Bioinformatics: Software Development and Applications

Lecture 10: AlphaFold2 for Protein Structure Prediction

Jie Zheng (郑杰)

PhD, Associate Professor

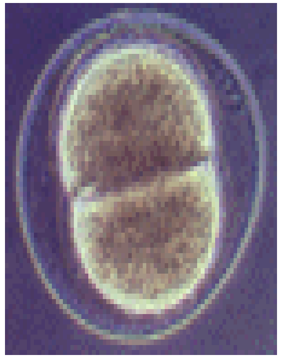
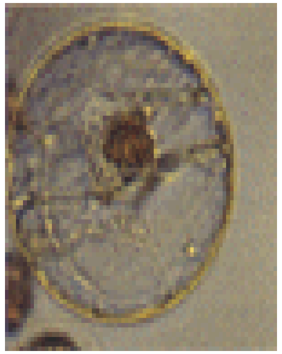
School of Information Science and Technology (SIST), ShanghaiTech University

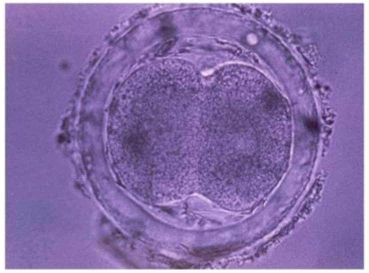
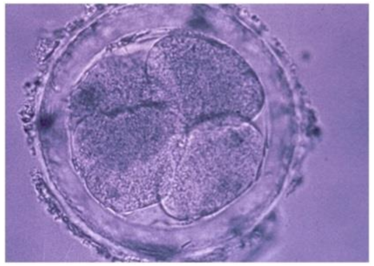
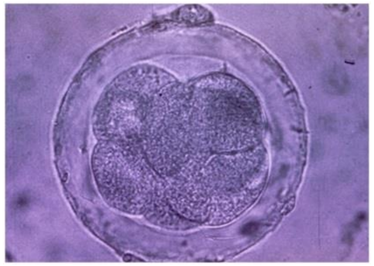
Spring, 2024

Outline

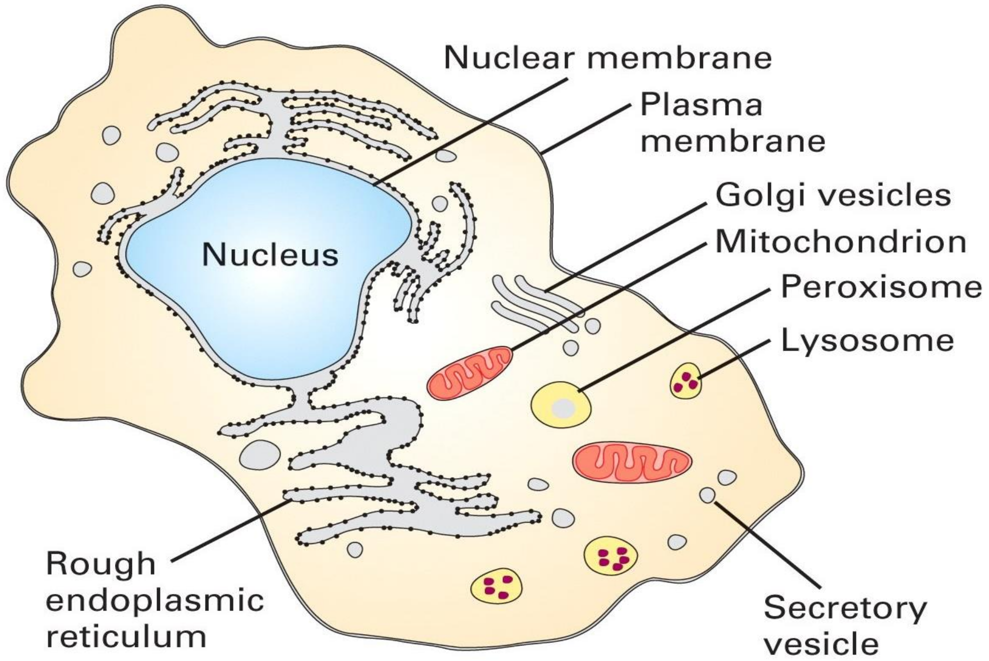
- Biological background
- AI for protein structure prediction using AlphaFold2 (AF2)
- Appendix: Applications of AF2, focused on drug discovery

What Is Life Made of?

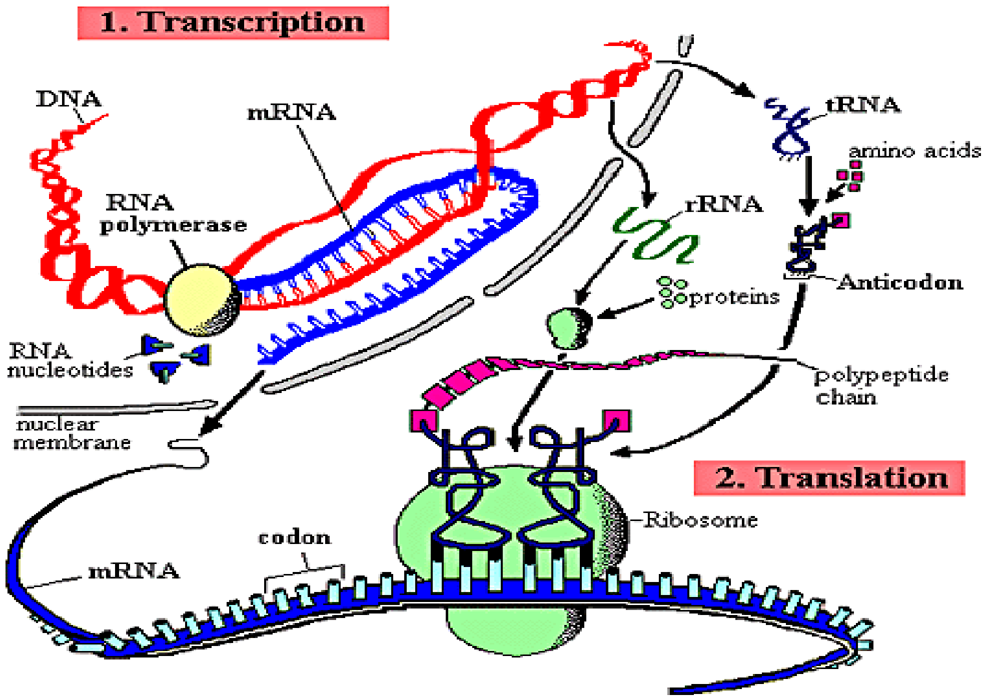




Life is made of cells

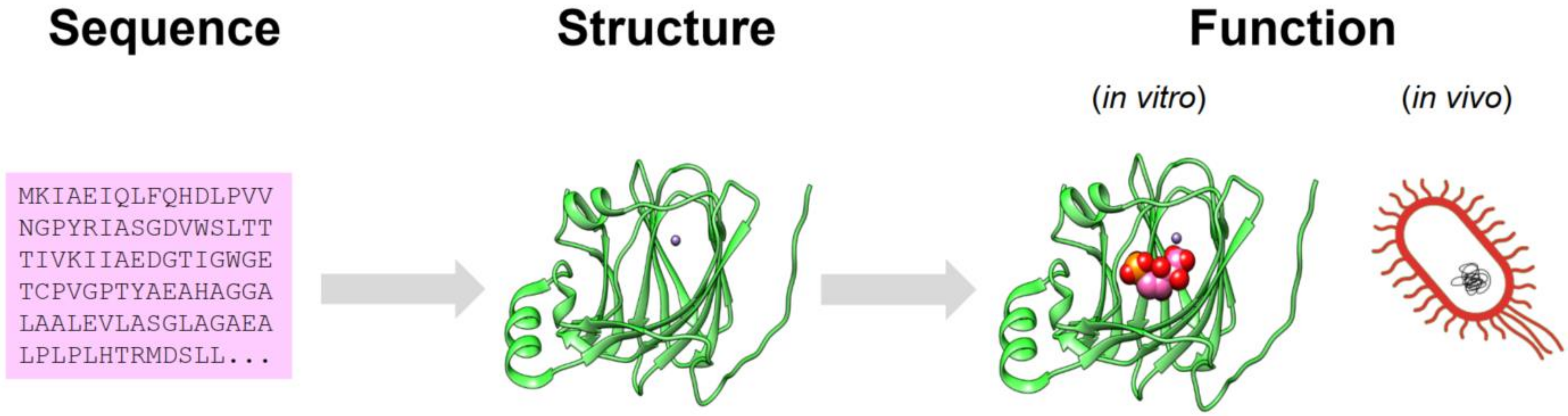


- A cell is a smallest structural unit of an organism that is capable of independent functioning
- All cells have some common features

The Central Dogma of Molecular Biology



- Genetic information flows from DNA to RNA to protein



Overview of protein structure

- The three-dimensional structure of a protein determines its capacity to function
- In 1950s, Christian B. Anfinsen and others denatured ribonuclease, observed rapid refolding, and demonstrated that the primary amino acid sequence determines its three-dimensional structure
- We can study protein structures to understand problems such as:
  - the consequence of disease-causing mutations
  - the properties of ligand-binding sites, and
  - the functions of genes, etc.

Christian B. Anfinsen (1916-1995)
American biochemist,
Nobel Prize in Chemistry (1972)

4 levels of protein structure

上海科技大学 ShanghaiTech University

- **Primary structure**: the linear sequence of amino acid residues in a polypeptide chain
- **Secondary structure**: the arrangements of the primary amino acid sequence into motifs such as α helices, β sheets and coils (or loops)
- **Tertiary structure**: the three-dimensional arrangement formed by packing secondary structure elements into globular domains
- **Quaternary structure**: the arrangement of several polypeptide chains, including interactions of one protein with other proteins, subunits and atoms

Protein primary and secondary structures

- Primary structure

MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKVKAHGKKVLGAFSD
GLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH

- Secondary structure

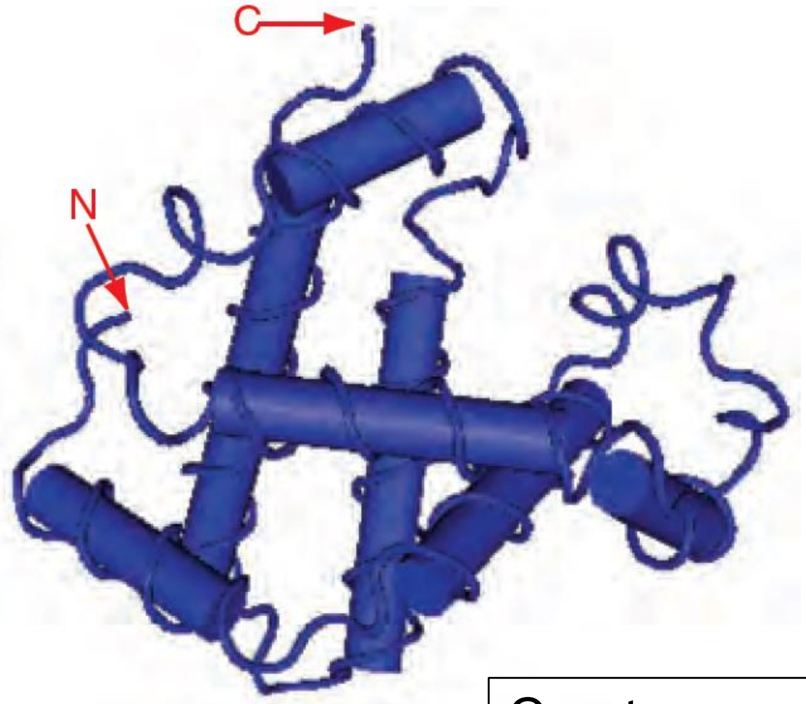
```
                    10        20        30        40        50        60        70
                     |         |         |         |         |         |         |
UNK_257900 MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKVKAHGKKVLG
DSC        cccchhhhhhhhhhhhcccccchhhhhhhhhhhccccchhhhhhhhccccccccccccchhhhhhhhhhh
MLRC       cccccchhhhhhhhhhccccccccchhhhhhheeecccchhhhhccccccccccccccccccccchhhhh
PHD        ccccchhhhhhhhhhhcccchhhcchhhhhhheeeccchhhhhhhhcccchhhhheccchhhhhhhhhhhh
Sec.Cons.  ccccchhhhhhhhhhhcccccchcchhhhhhheeeccchhhhhhhhccccccccccccchhhhhhhhhhh

                    80        90       100       110       120       130       140
                     |         |         |         |         |         |         |
UNK_257900 AFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVAN
DSC        hhhhhhhhhhhhhhhhhhhhhhhhhhccccchhhhhhhhhhhhhhhhhccccccchhhhhhhhhhhhhhh
MLRC       hhhhhhhhhhhhhhhhhhhhhhhcccccccccchhhhhhhhhhhhhhhhcccccchhhhhhhhhhhhhhhh
PHD        hhhhhhhhhhhhhhhhhhhhhhhhhccccchhhhhhhhhhhhhhhhhhhccccchhhhhhhhhhhhhhhh
Sec.Cons.  hhhhhhhhhhhhhhhhhhhhhhhhhcccccchhhhhhhhhhhhhhhhhcccccchhhhhhhhhhhhhhhh

UNK_257900 ALAHKYH
DSC        hhhhccc
MLRC       hhhhccc
PHD        hhhhhcc
Sec.Cons.  hhhhccc
```

Results from three secondary structure programs are shown, with their consensus:

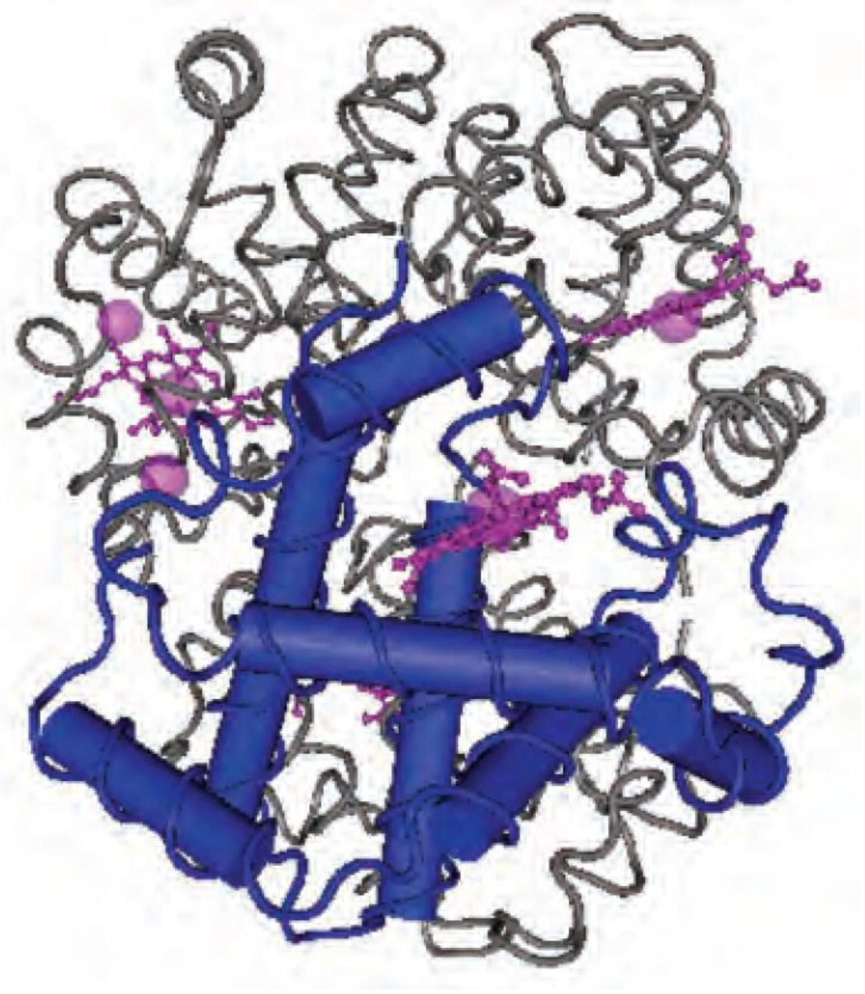
- **h**: alpha helix
- **c**: random coil
- **e**: extended strand

Protein tertiary and quaternary structures

Tertiary structure



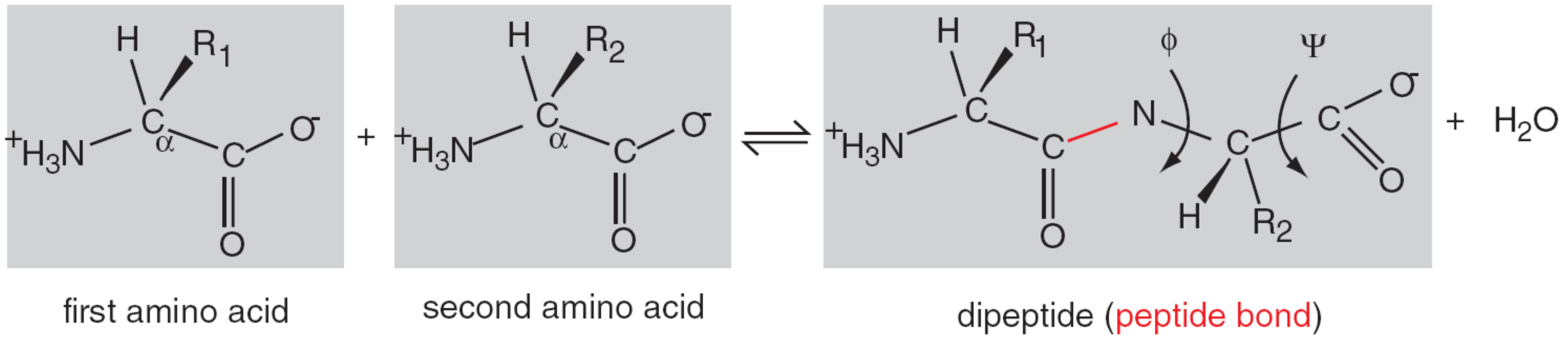
Quarternary structure



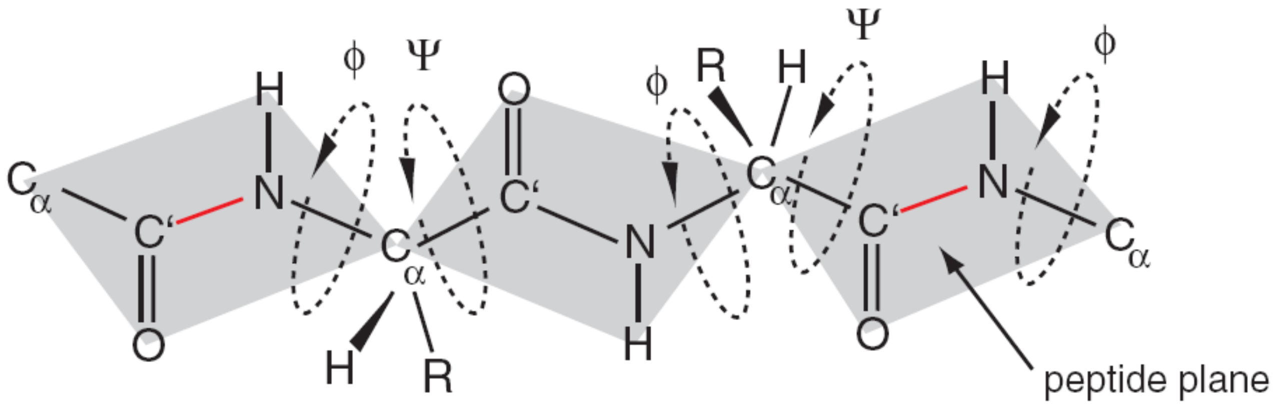
Quarternary structure: the four subunits of hemoglobin are shown (with one beta globin chain highlighted) as well as four noncovalently attached heme groups.

Peptide bond: phi and psi angles

(a) peptide bond



(b) phi and psi angles of polypeptide



Tertiary protein structure: protein folding

Experimental determination

- X-ray crystallography
- Nuclear magnetic resonance (NMR)
- Cryogenic electron microscopy (cryo-EM)

Computational prediction

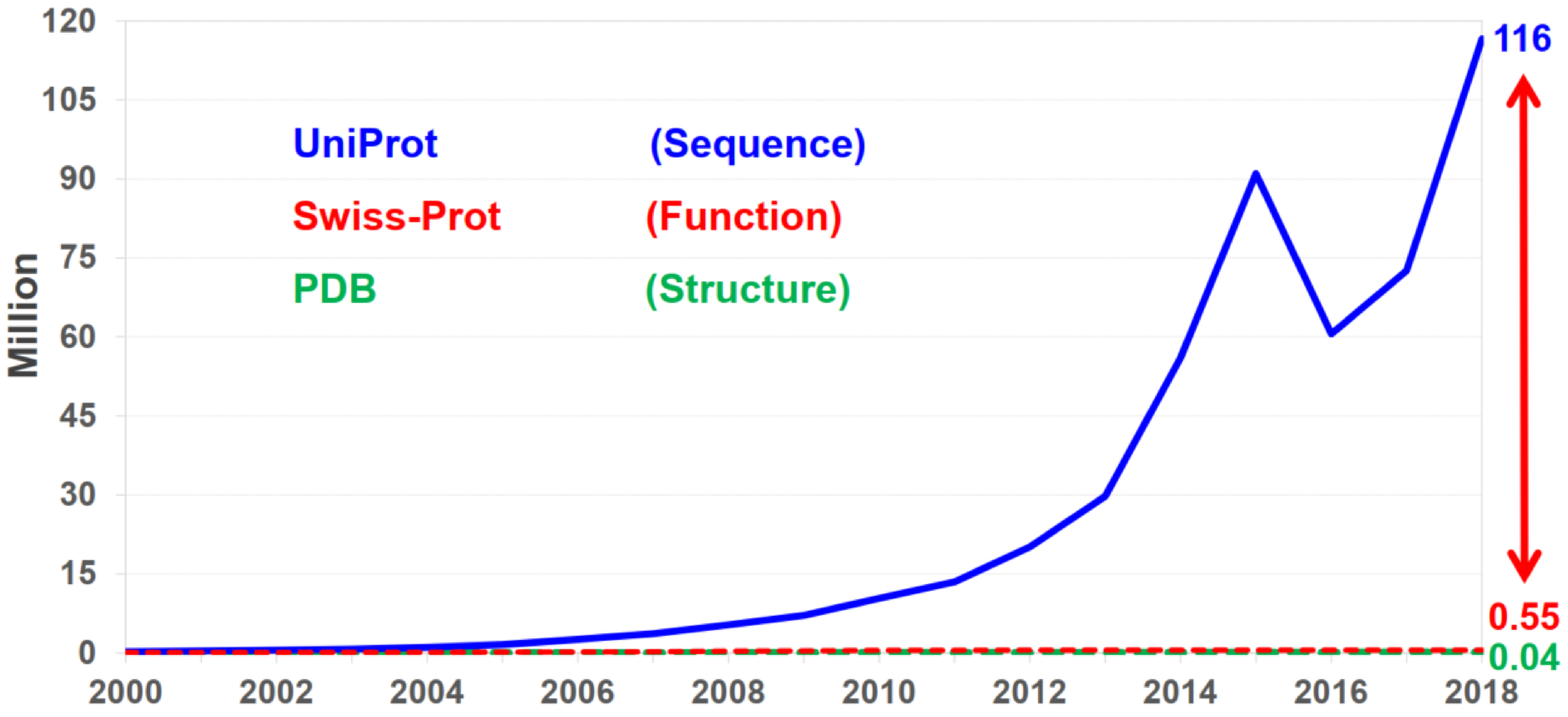
- Homology modeling (comparative modeling)
- Fold recognition (threading)
- *Ab initio* prediction (template-free modeling)

AI for Protein 3D Structure Prediction

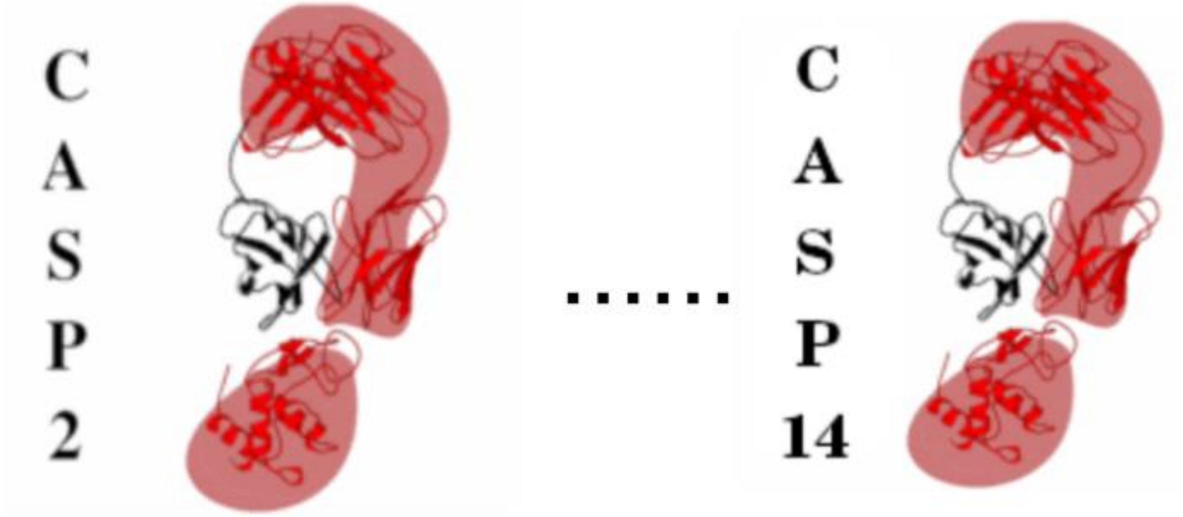
上海科技大学
ShanghaiTech University

立志 成才 报国 裕民

Sequence-structure/function gap keeps increasing: need computational prediction



The Critical Assessment of protein Structure Prediction (CASP)



1994 1996 2020

CASP1	1994
CASP2	1996
CASP3	1998
CASP4	2000
CASP5	2002
CASP6	2004
CASP7	2006
CASP8	2008
CASP9	2010
CASP10	2012
CASP11	2014
CASP12	**2016**
CASP13	2018
CASP14	2020

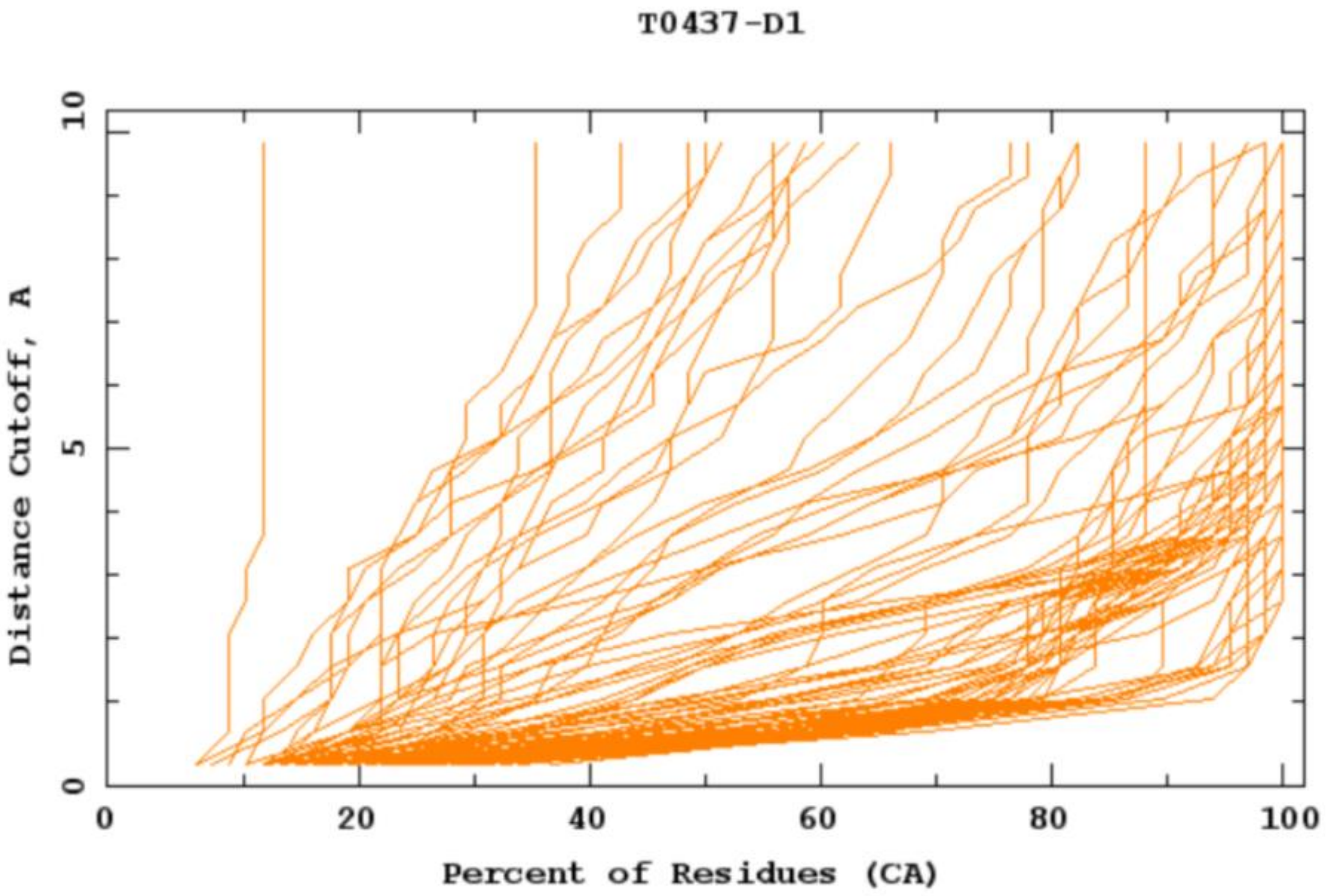
http://www.predictioncenter.org/

Evaluation of prediction by GDT_TS

GDT_TS = (GDT_P1 + GDT_P2 + GDT_P4 + GDT_P8)/4

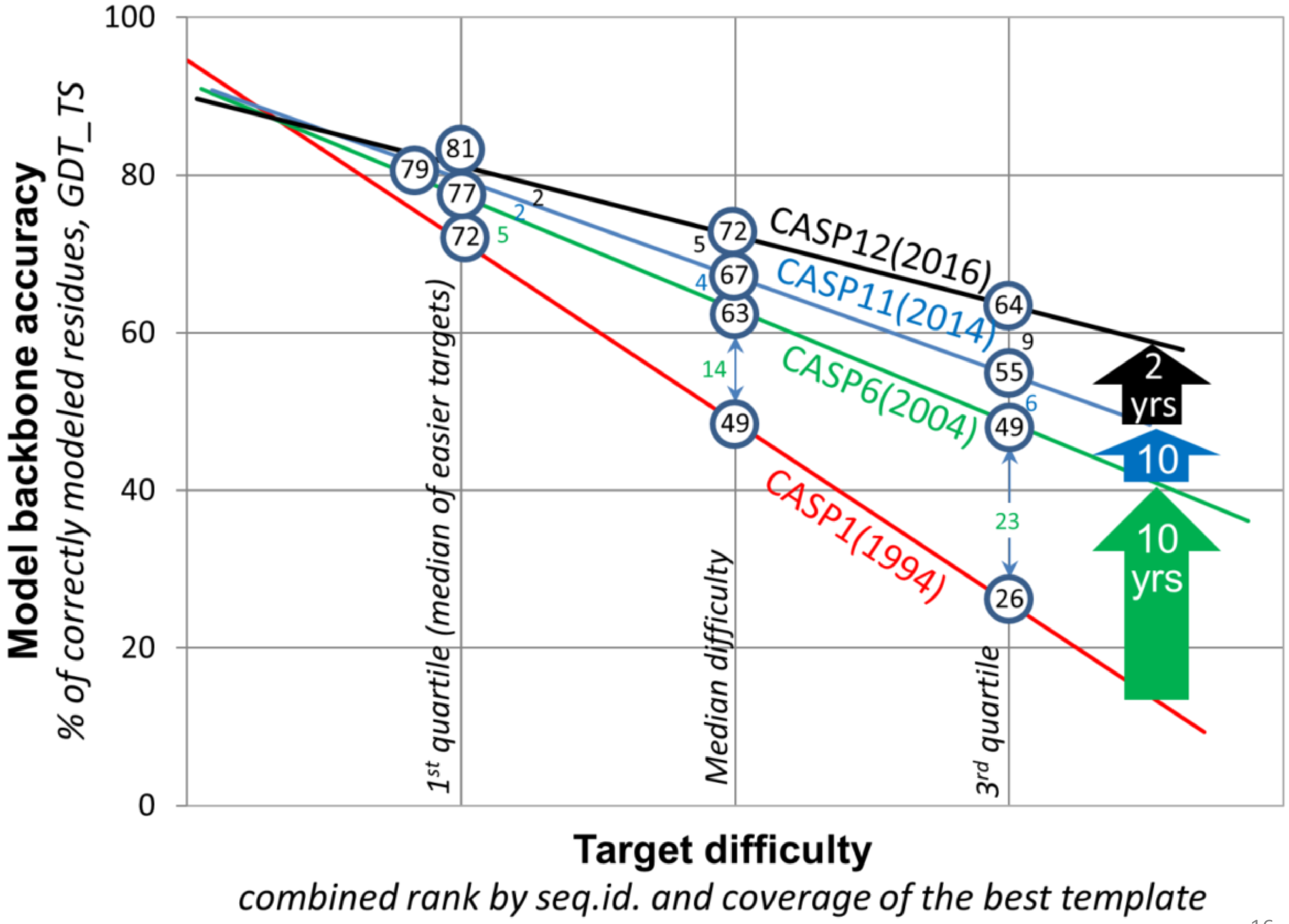
GDT_TS: GlobalDistanceTest_TotalScore

where GDT_Pn denotes percent of residues under distance cutoff <= nÅ



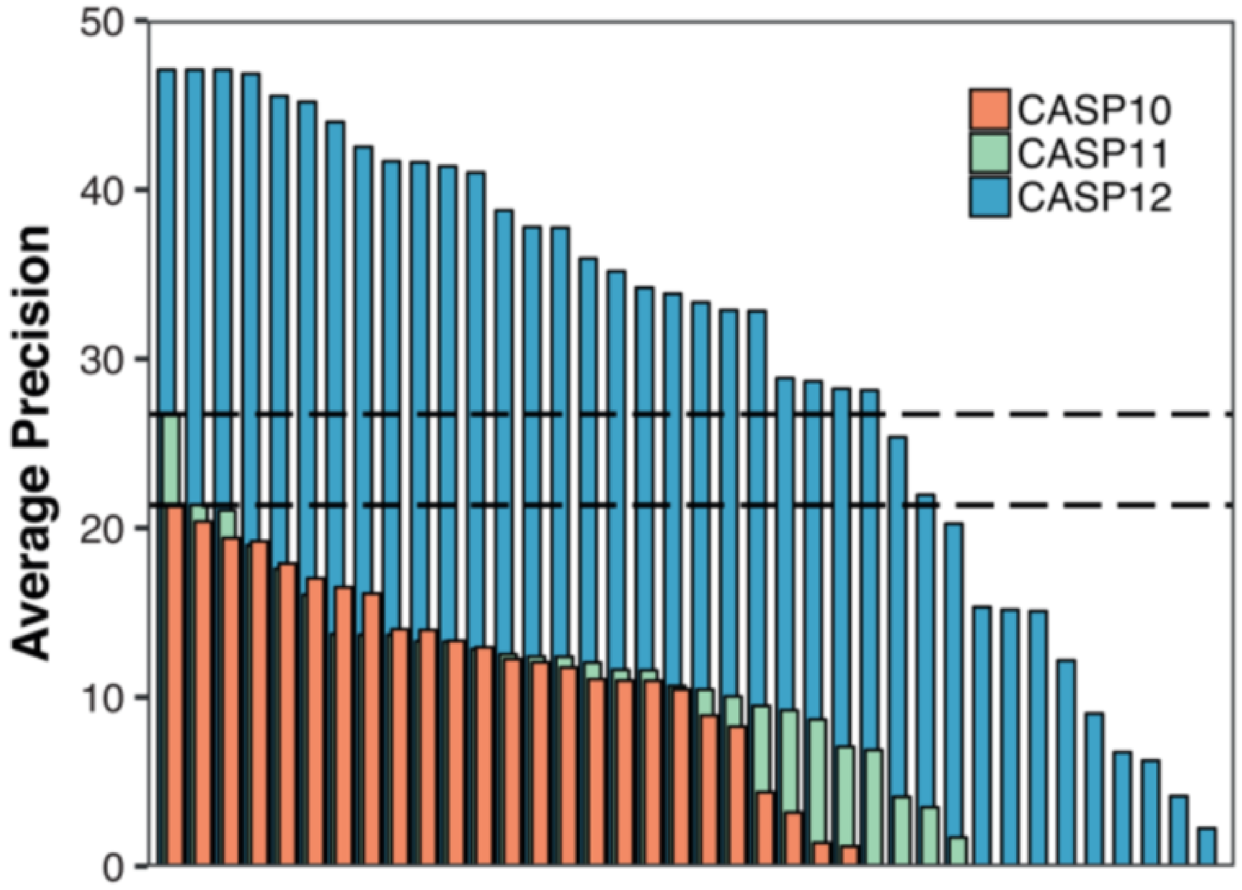
蛋白结构
预测曾经
十年停滞

2016年又
重新爆发



Revolution starts from CASP12 (2016)

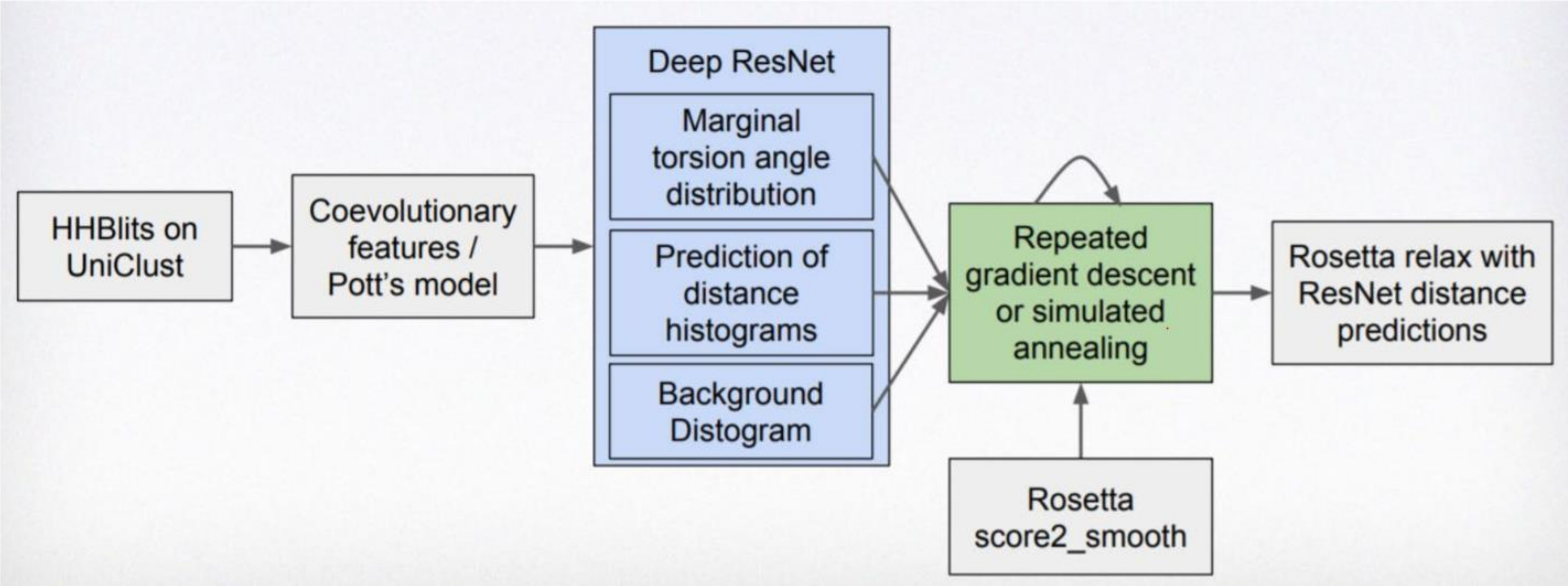
Contact map prediction represented by RaptorX-Contact



许锦波

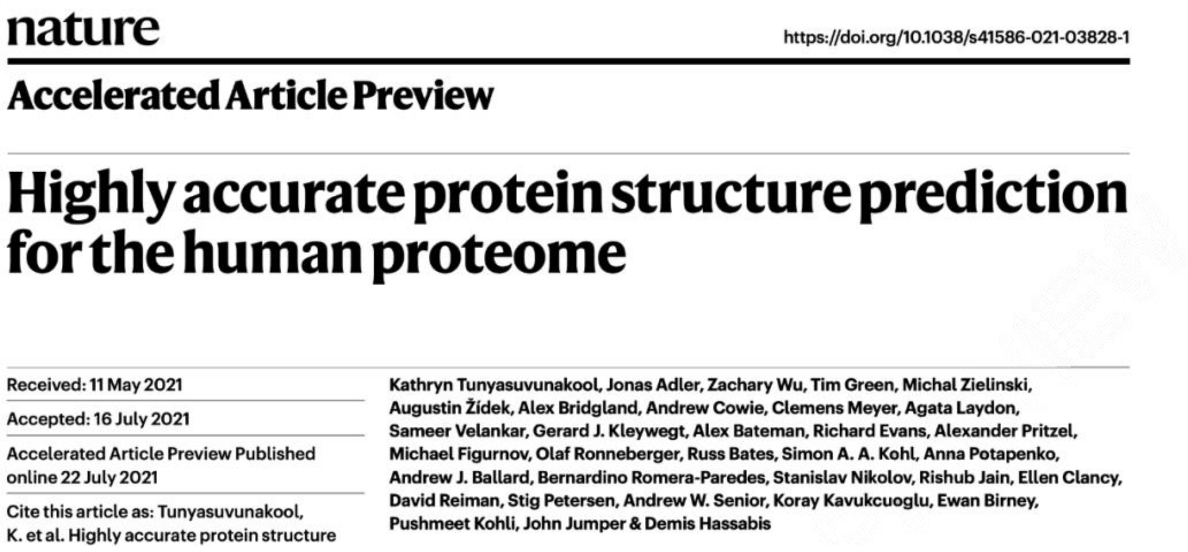
清华大学卓越访问教授、美国芝加哥丰田计算技术研究所教授、分子之心创始人

No. 1 of CASP13: AlphaFold



AlphaFold2 高精度破译人类蛋白质结构

上海科技大学 ShanghaiTech University

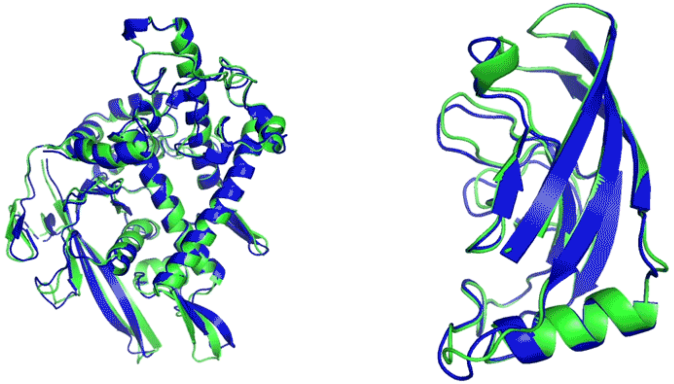


nature
https://doi.org/10.1038/s41586-021-03828-1
Accelerated Article Preview

Highly accurate protein structure prediction for the human proteome

Received: 11 May 2021
Accepted: 16 July 2021
Accelerated Article Preview Published online 22 July 2021
Cite this article as: Tunyasuvunakool, K. et al. Highly accurate protein structure

Kathryn Tunyasuvunakool, Jonas Adler, Zachary Wu, Tim Green, Michal Zielinski, Augustin Žídek, Alex Bridgland, Andrew Cowie, Clemens Meyer, Agata Laydon, Sameer Velankar, Gerard J. Kleywegt, Alex Bateman, Richard Evans, Alexander Pritzel, Michael Figurnov, Olaf Ronneberger, Russ Bates, Simon A. A. Kohl, Anna Potapenko, Andrew J. Ballard, Bernardino Romera-Paredes, Stanislav Nikolov, Rishub Jain, Ellen Clancy, David Reiman, Stig Petersen, Andrew W. Senior, Koray Kavukcuoglu, Ewan Birney, Pushmeet Kohli, John Jumper & Demis Hassabis



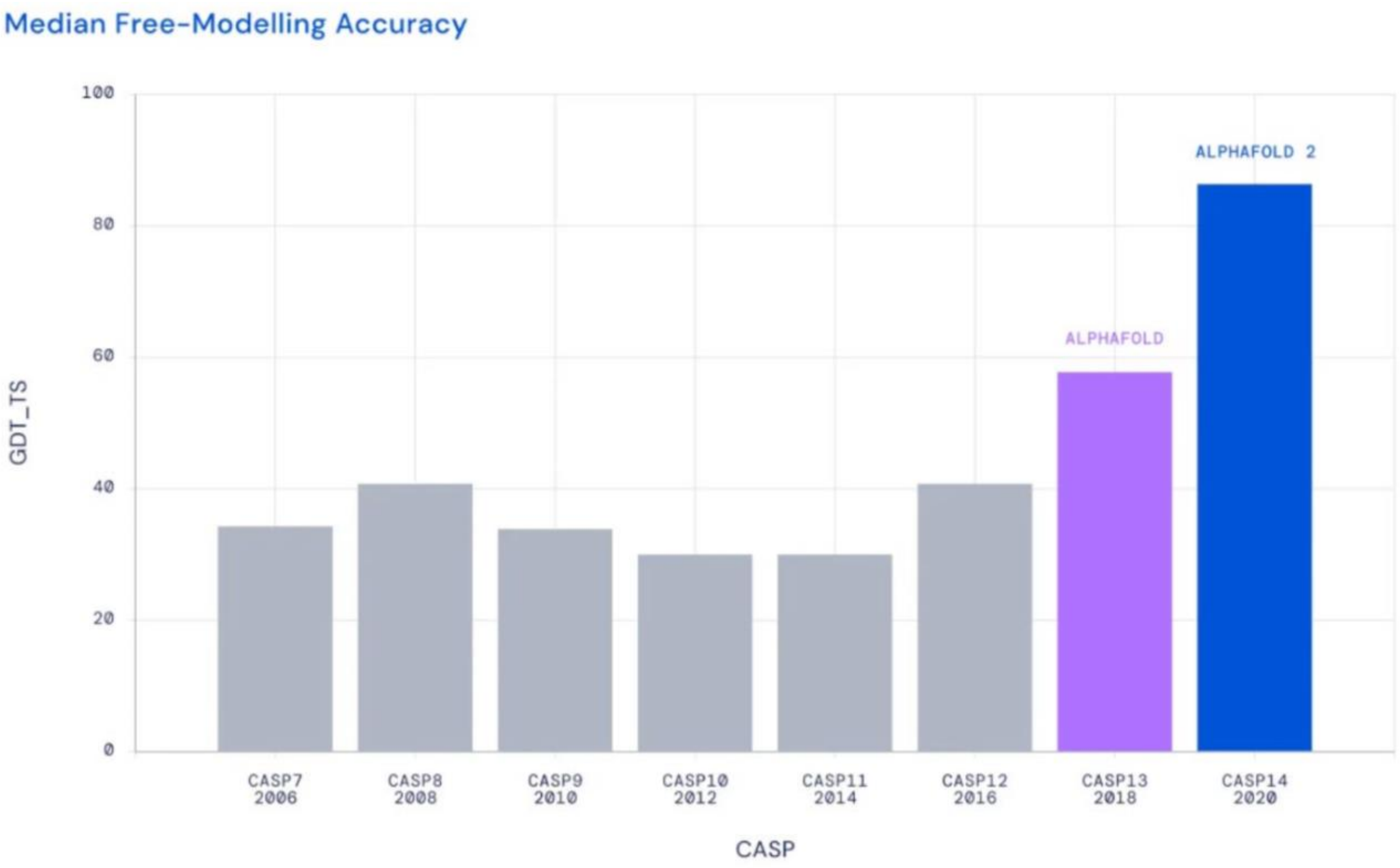
2020年，DeepMind的AlphaFold2 (AF2) 在CASP14中取得了碾压性成绩：

- **大幅提高全原子蛋白结构预测的置信度和覆盖率**
- **蛋白结构的准确预测带来了高质量的生物学假设**
- **AI 驱动生物学研究的时代开始了**

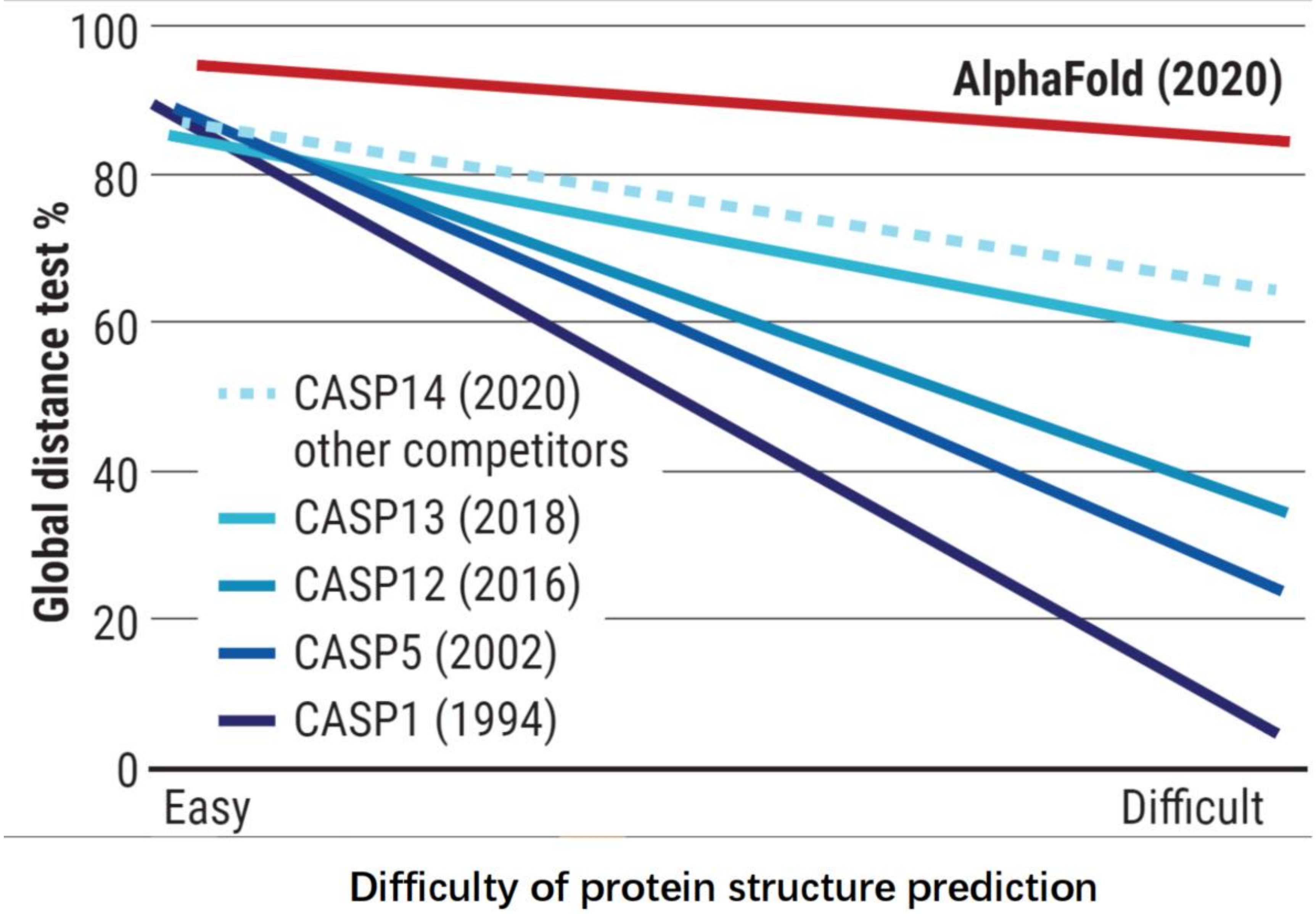
CASP14 AF2预测样例展示：
绿色为真实结构，蓝色为预测结构。
GDT分数大于90（满分100）可以认为和真实结构差别不大。

立志 成才 报国 裕民

AlphaFold2’s accuracy is near experimental measurement

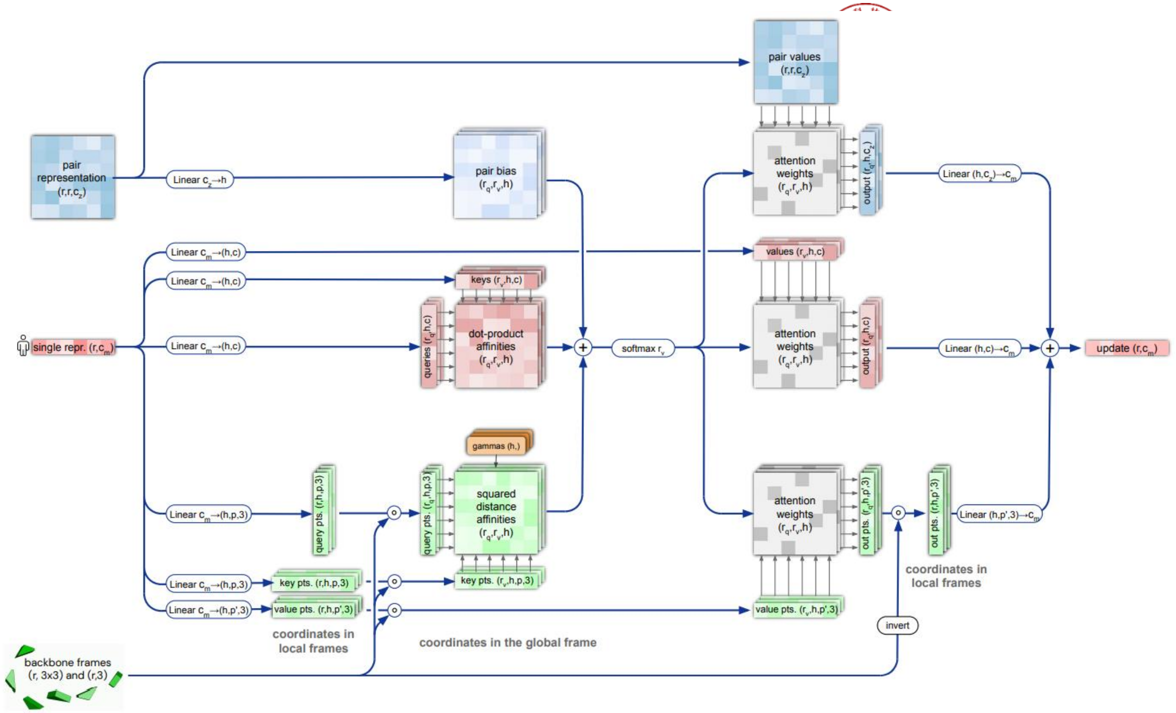


AF2 performs well even for difficult proteins



AlphaFold2 paper

- From Google DeepMind
- "Highly accurate protein structure prediction with AlphaFold" (Jumper et al. Nature 2021)
- Comprehensive description of AlphaFold2
  - 60 pages of supplementary materials
  - 32 algorithms
- Alongside it DeepMind open sourced the code:
  - Code: github.com/deepmind/alphafold
  - Colab: dpmd.ai/alphafold-colab



Algorithm 31 Generic recycling training procedure

def RecyclingTraining$(\{\text{inputs}_c\}, N_\text{cycle})$:

$N' = \text{uniform}(1, N_\text{cycle})$ *# shared value across the batch*

$\text{outputs} = \mathbf{0}$

 # Recycling iterations

for all $c \in [1, \ldots, N']$ **do** *# shared weights*

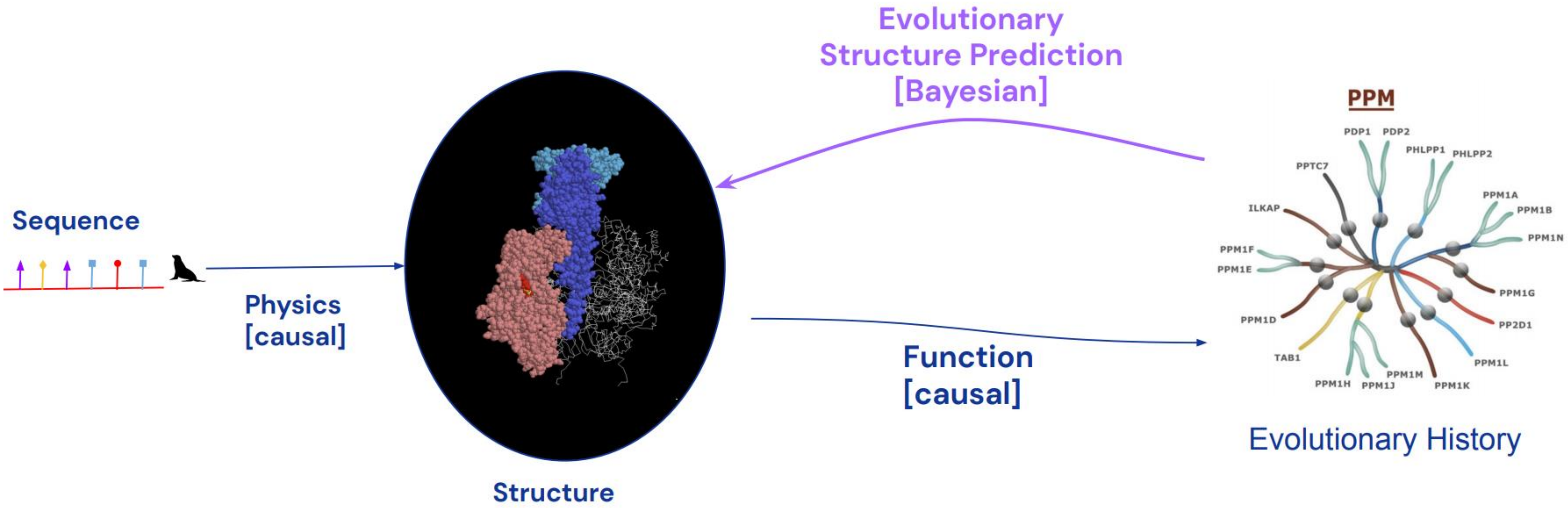
  $\text{outputs} \leftarrow \text{stopgrad}(\text{outputs})$ *# no gradients between iterations*

  $\text{outputs} \leftarrow \text{Model}(\text{inputs}_c, \text{outputs})$

end for

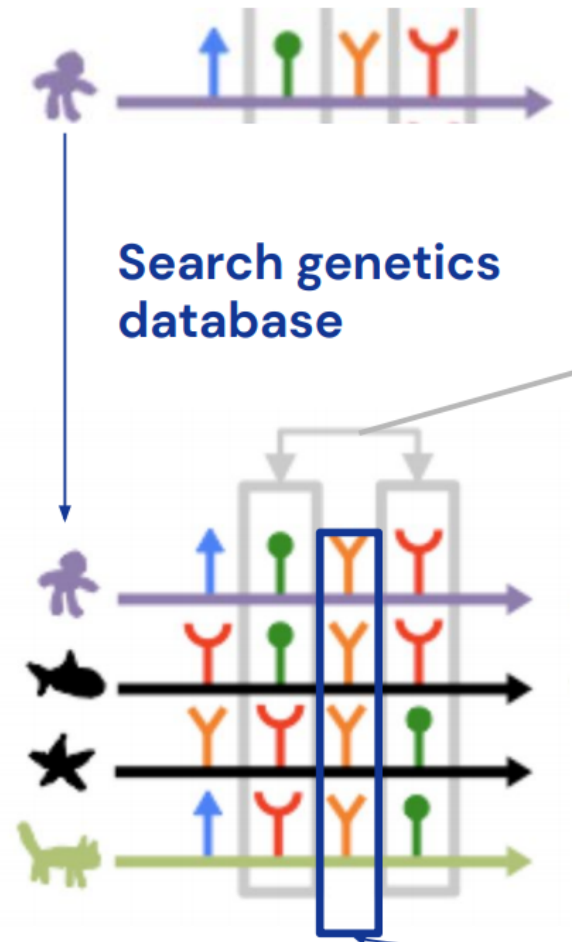
return loss(outputs) *# only the final iteration's outputs are used*

Determining structure from evolution - Intuition

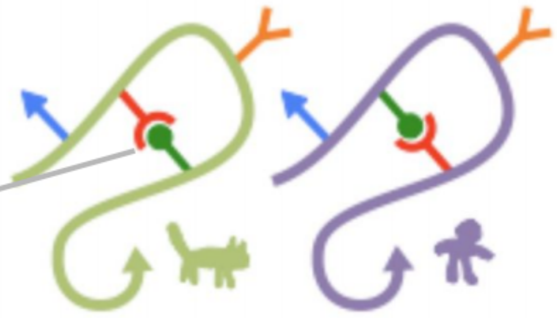


Source of slides: Google DeepMind

Determining structure from evolution – Intuition (continued)



Multiple Sequence Alignment (MSA)



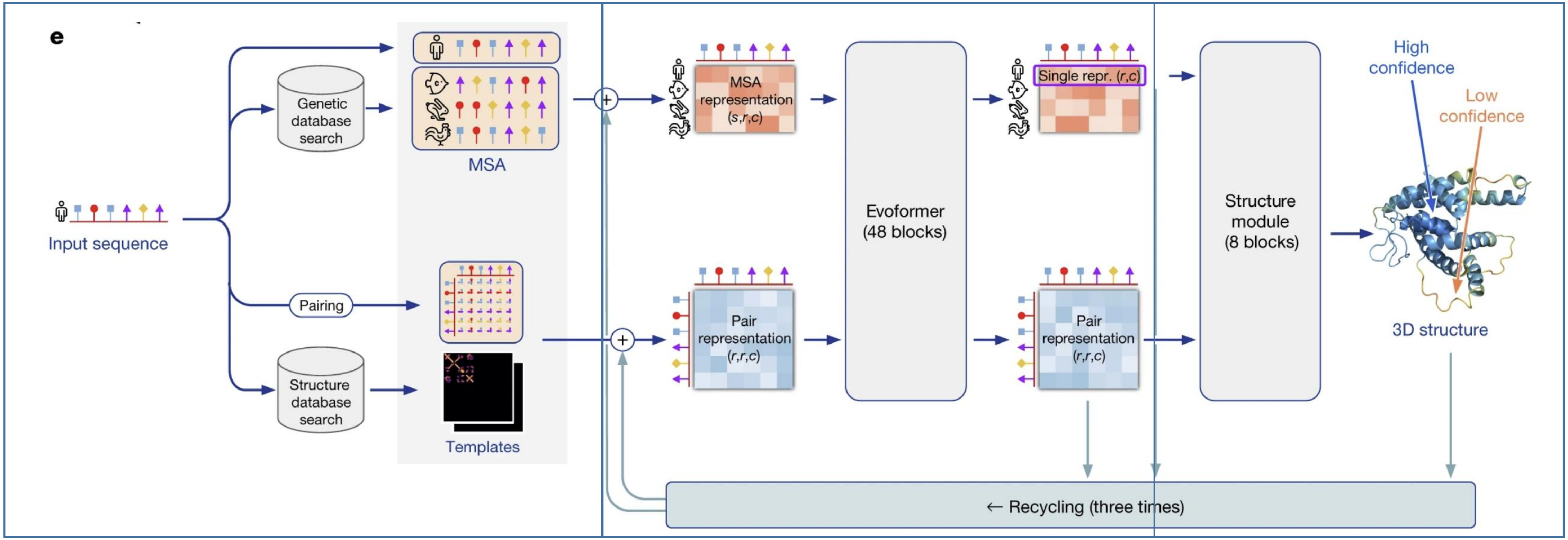
Co-evolution: residues in contact tend to mutate together.

(mutation of a single residue breaks the contact and the organism with the mutated protein does not survive)

Evolution conserves some properties like hydrophobic/hydrophilic amino acids on the "inside"/"outside"

Coevolution cartoon by Sergey Ovchinnikov (https://jgi.doe.gov/seeking-structure-metagenome-sequences/cartoon-coevolution-sergey-o/)

AlphaFold2的模型结构

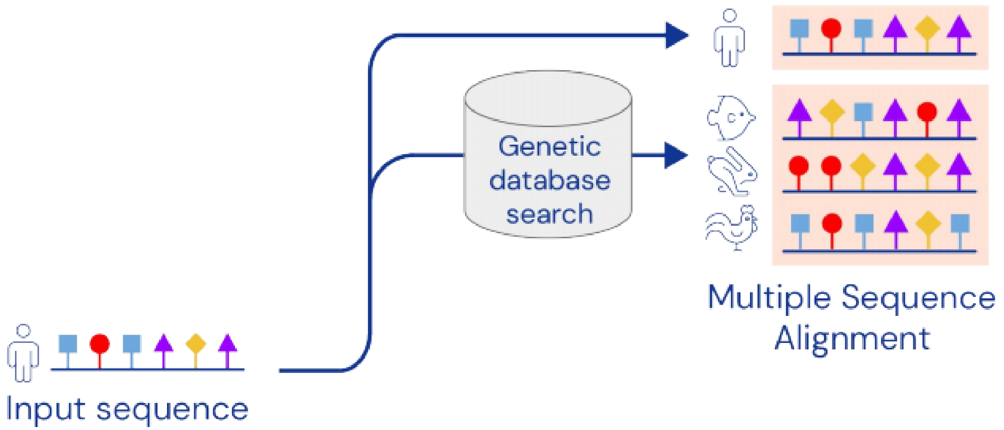


AF2架构可以分为3个模块：

1. **Feature Embedder**：用于处理输入的蛋白序列，MSA和结构模板
2. **共进化蛋白语言模型**：称为Evoformer（Evolutionary Transformer）
3. **结构生成**：本质上是一个3D等变Transformer

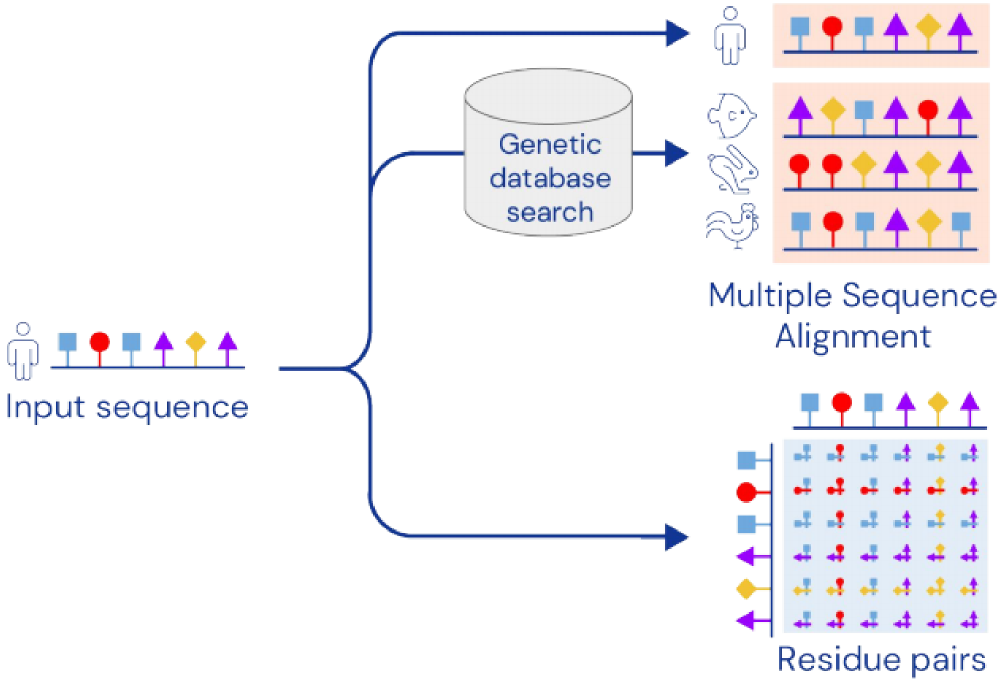
立志成才报国裕民

Inputs



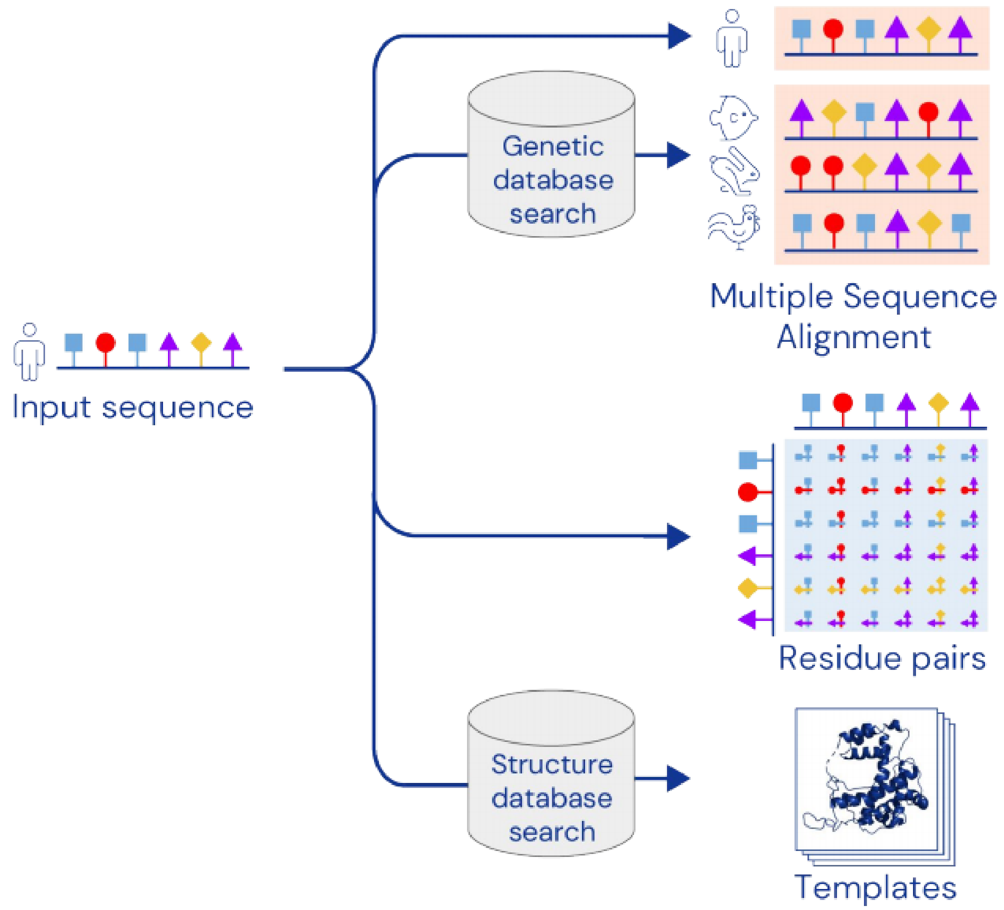
A key AlphaFold input is the MSA, containing sequences evolutionarily related to the target. Related sequences are found using standard tools and public databases.

Inputs



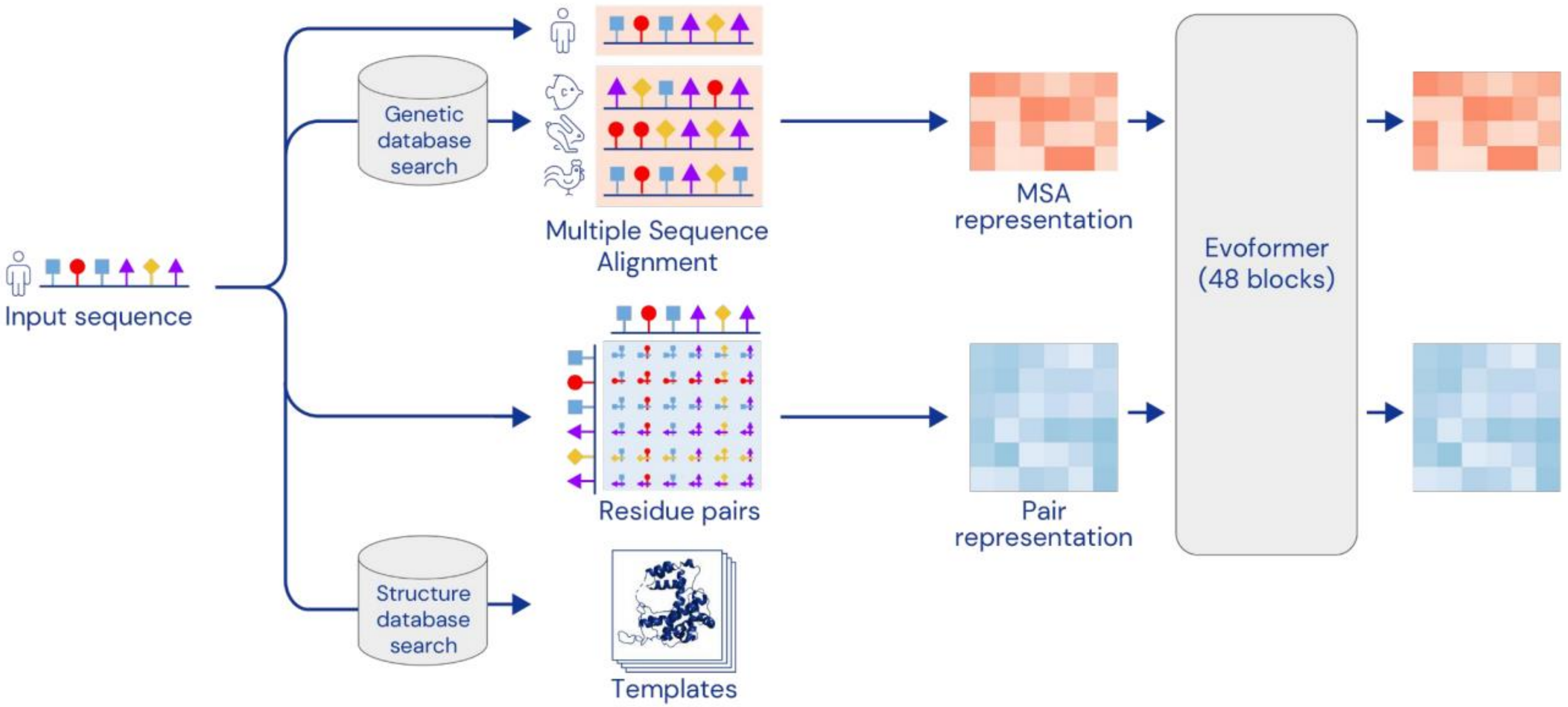
The input sequence is used to create an array of representations representing all residue pairs.

Inputs



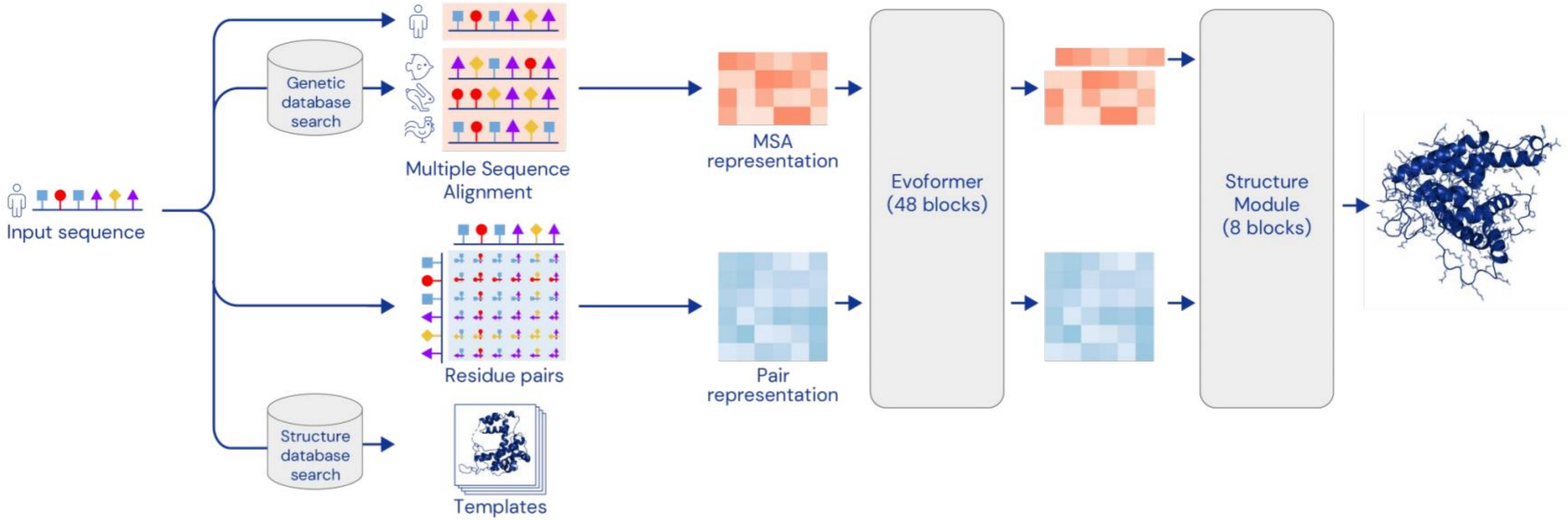
AlphaFold can also use template structures from the PDB, found using standard tools.
However, it often produces accurate predictions without a template.

Network



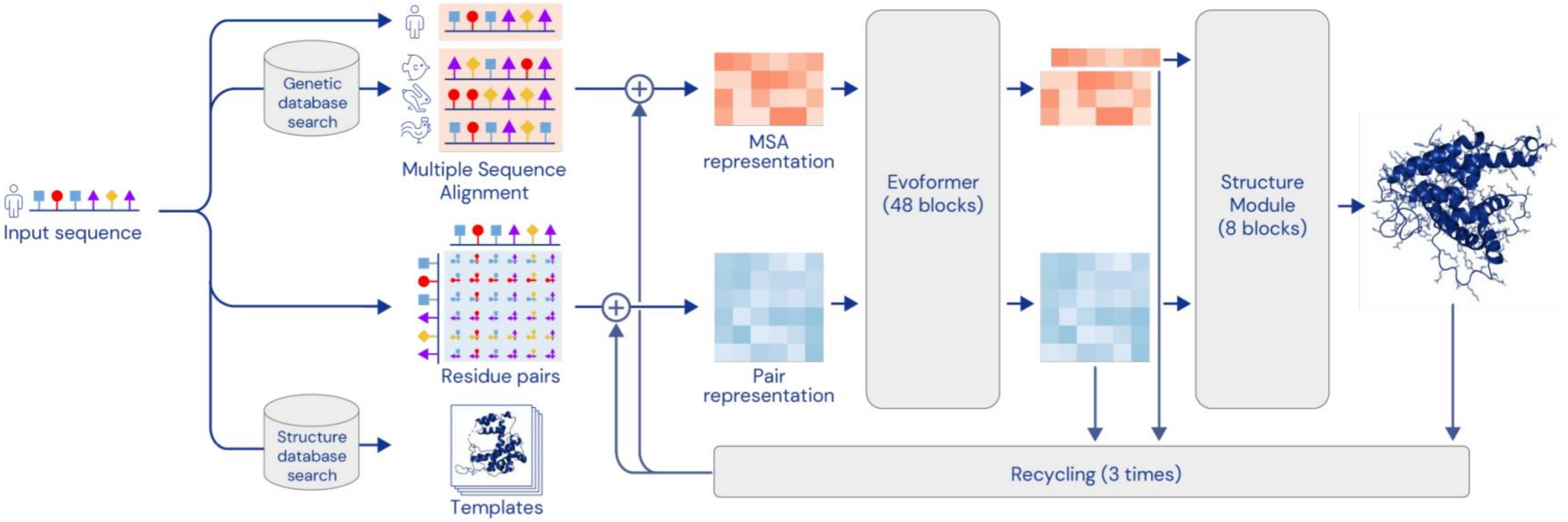
The Evoformer blocks extract information about the relationship between residues.
The MSA representation can update the pair representation and vice versa.

Network



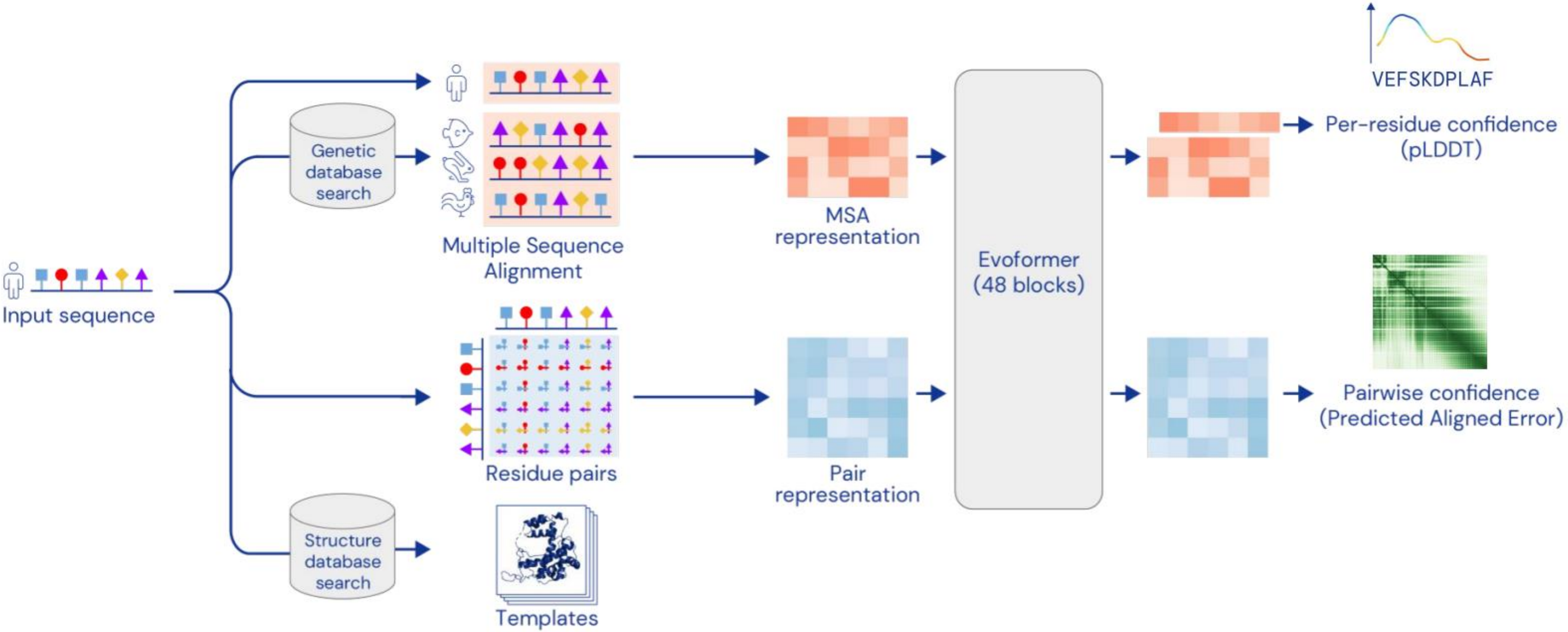
The Structure Module predicts a rotation + translation to place each residue.
A small network predicts side chain chi angles.

Network



Feeding certain outputs back through the network again improves performance

Confidence estimates

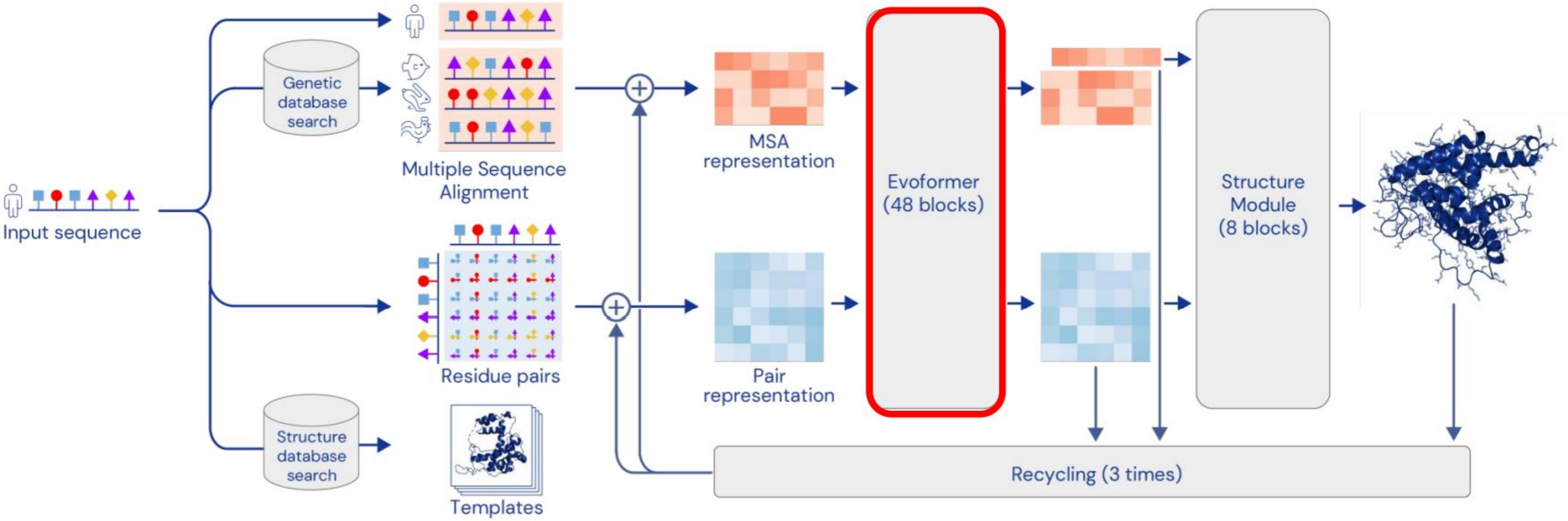


AlphaFold is trained to predict its confidence via simple supervision on the training set

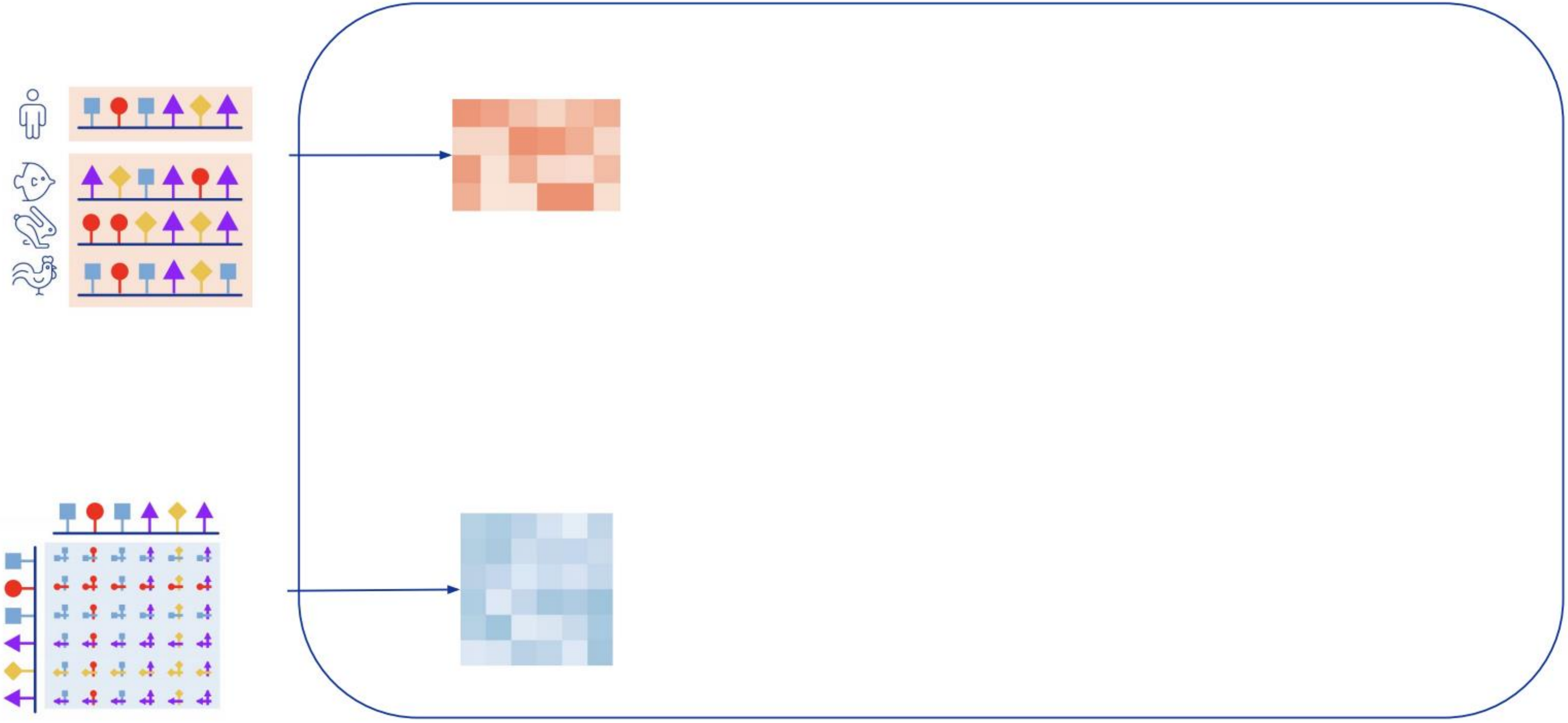
We use these confidences for ranking of predictions

Evoformer

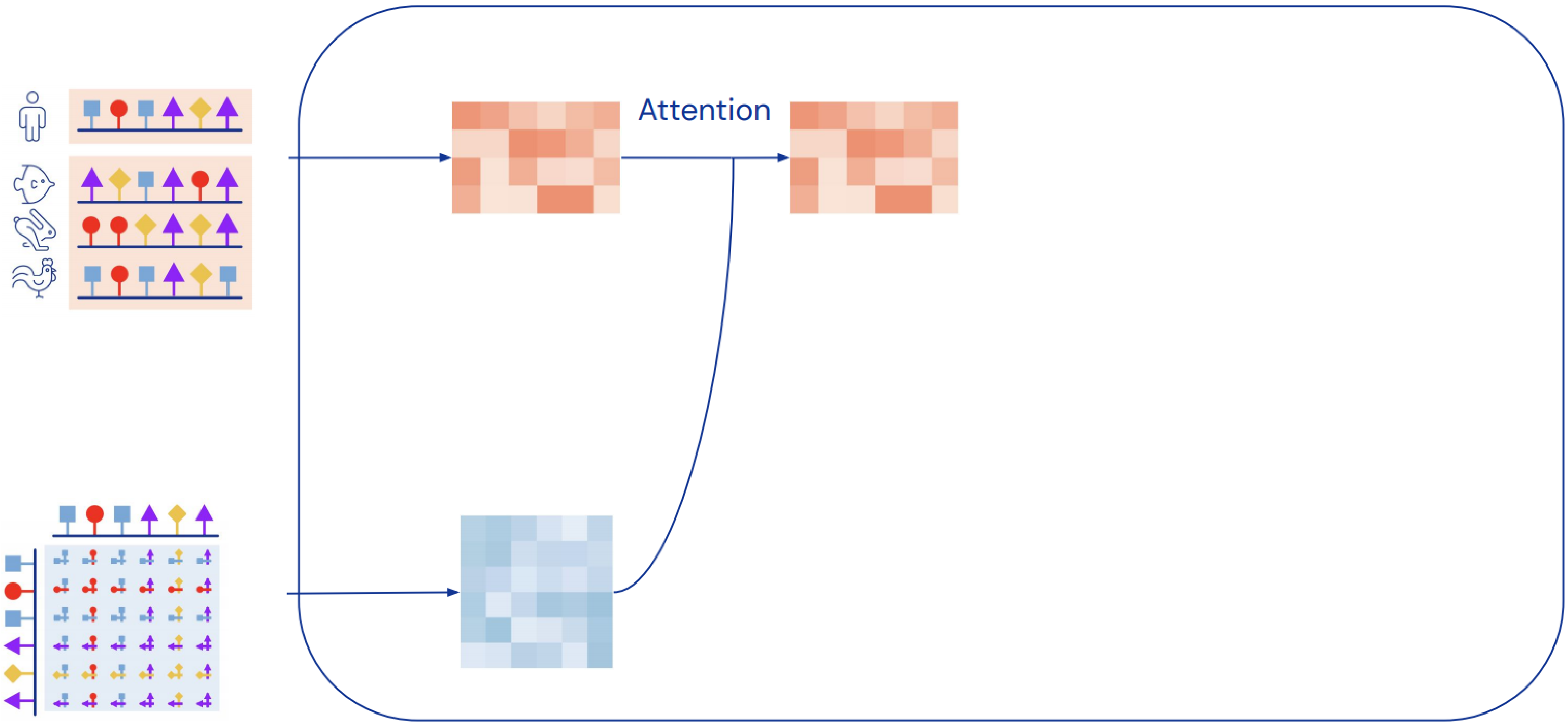
Network


Input sequence
Genetic database search
Multiple Sequence Alignment
Residue pairs
Structure database search
Templates
MSA representation
Pair representation
Evoformer (48 blocks)
Structure Module (8 blocks)
Recycling (3 times)

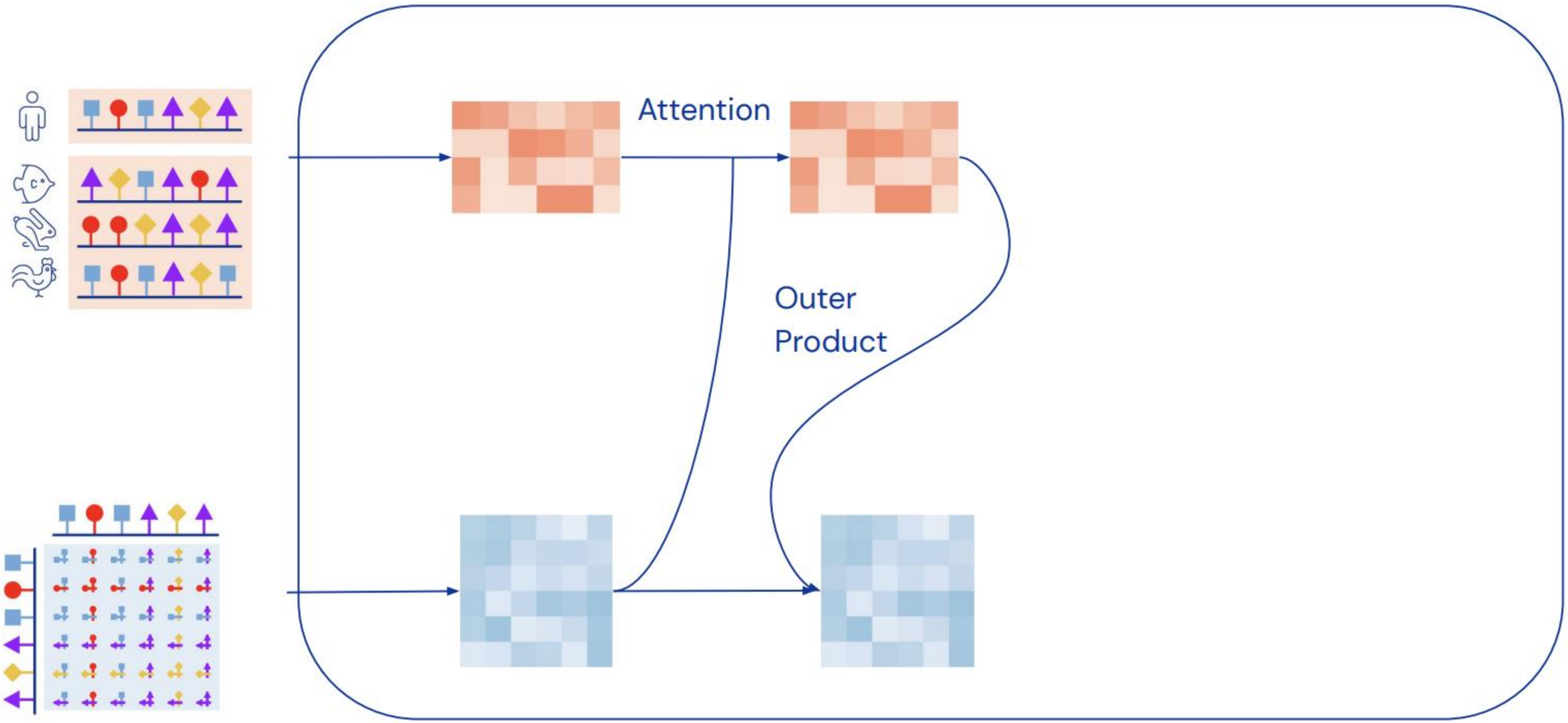
Evoformer



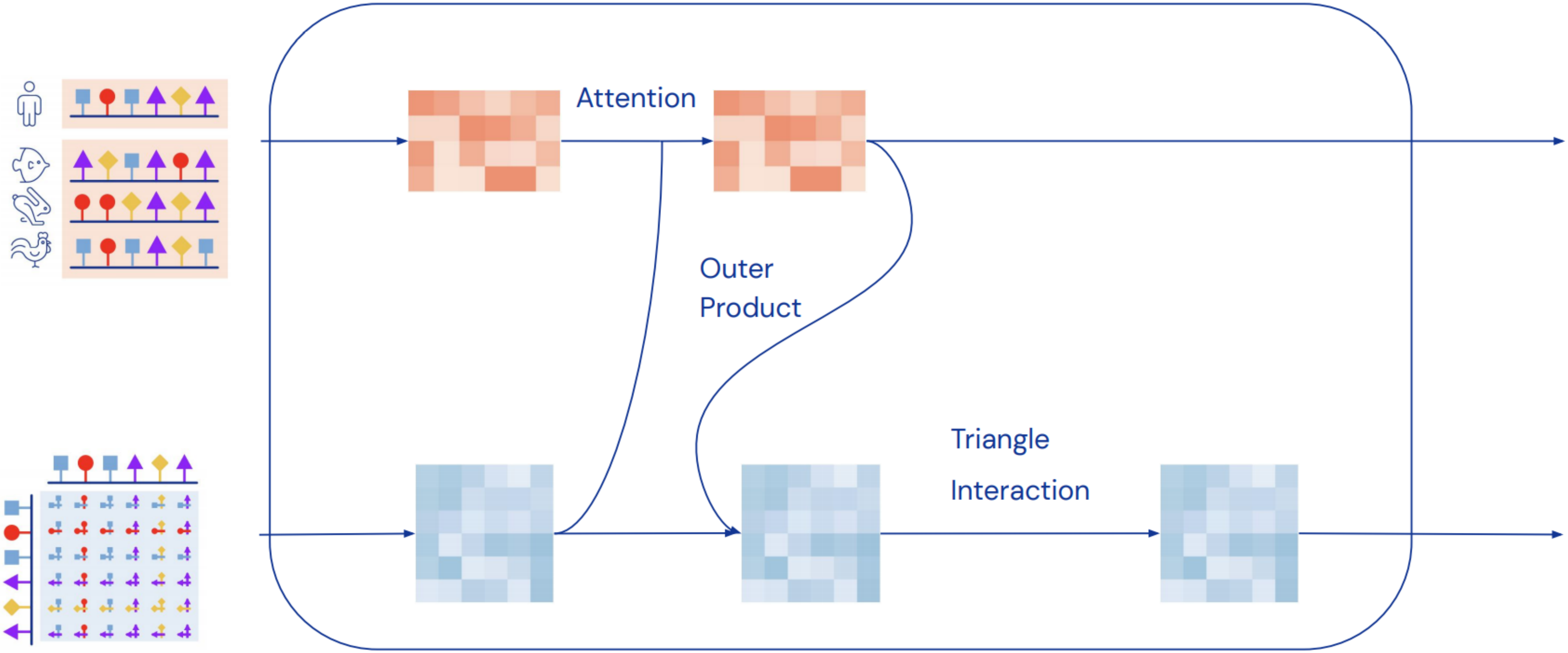
Evoformer



Evoformer

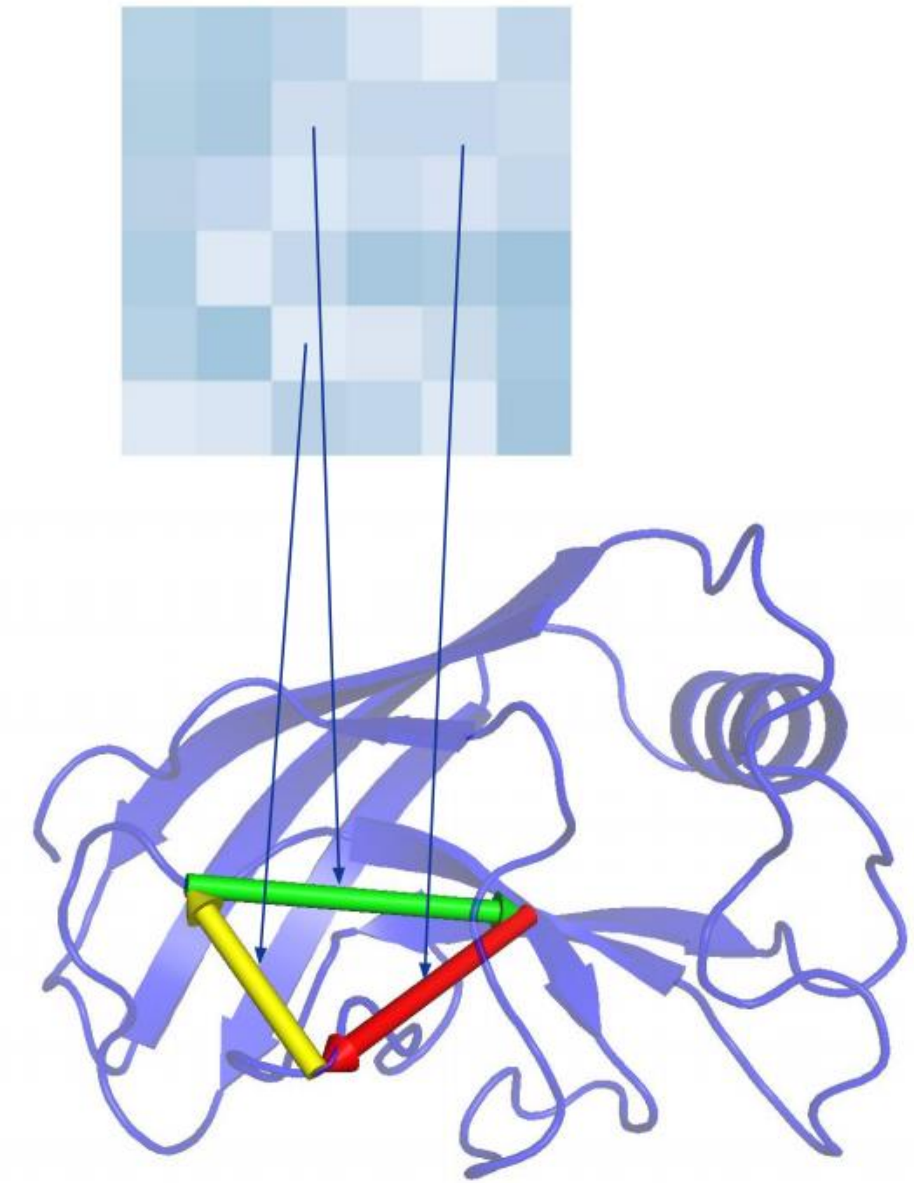


Evoformer



Triangle interaction

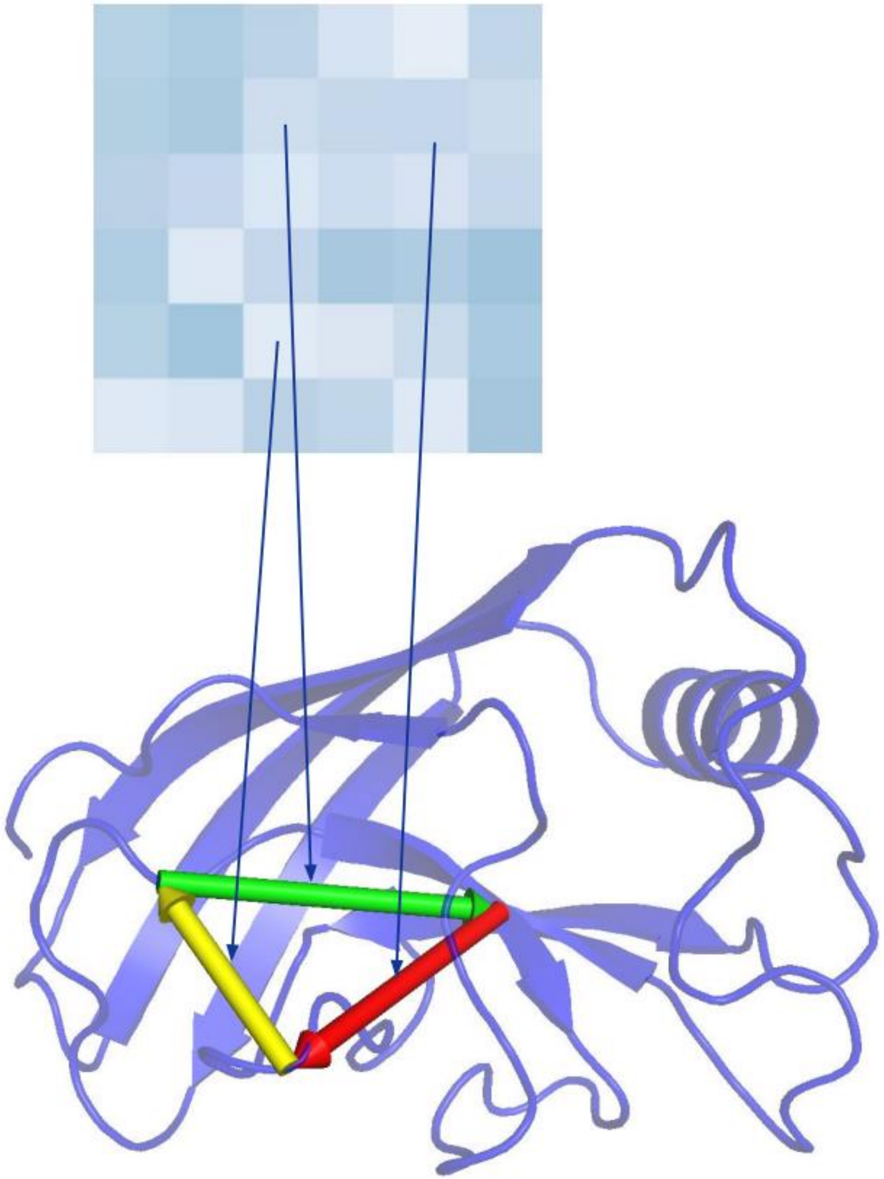
- Take 3 points A, B, C
  - If Distance AB and distance BC known strong constraint on AC (triangle inequality)
  - Evolution & Sequence gives information about relations between residues
- Pair Embedding encodes relations
  - Update for pair AC should depend on BC, AB



立志成才报国裕民

Graph inference

- ➔ **Graph**
  - ◆ Edges = Pairs of residues
  - ◆ We don't know the graph, need to infer it
  - ◆ Triplet relation in this language: length 3 cycles in graph
  - ◆ Update ╱ based on all cycles involving edge
- ➔ **Transitivity inductive bias**
  - ◆ This encodes transitivity bias of relations (E.g. triangle inequality, Loop Closure)



Evoformer

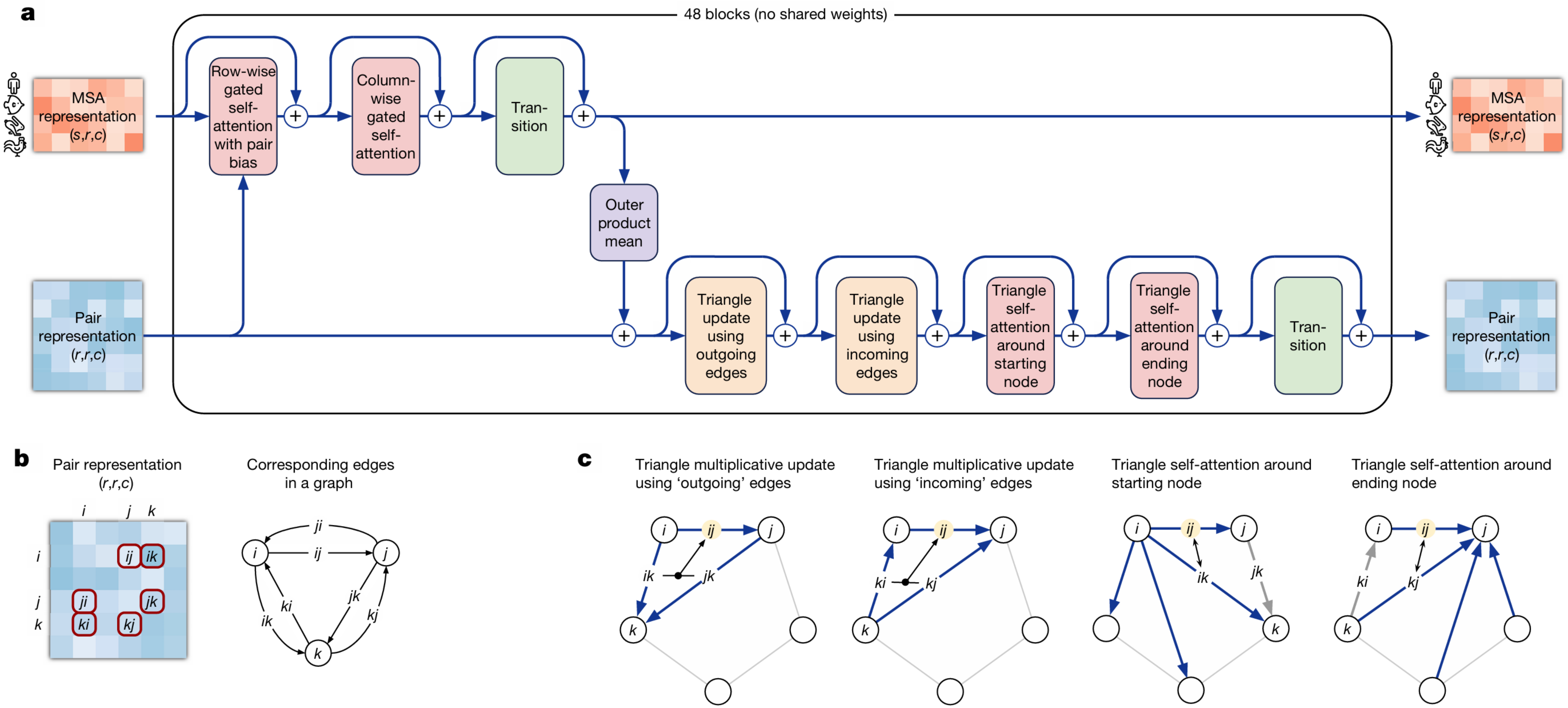
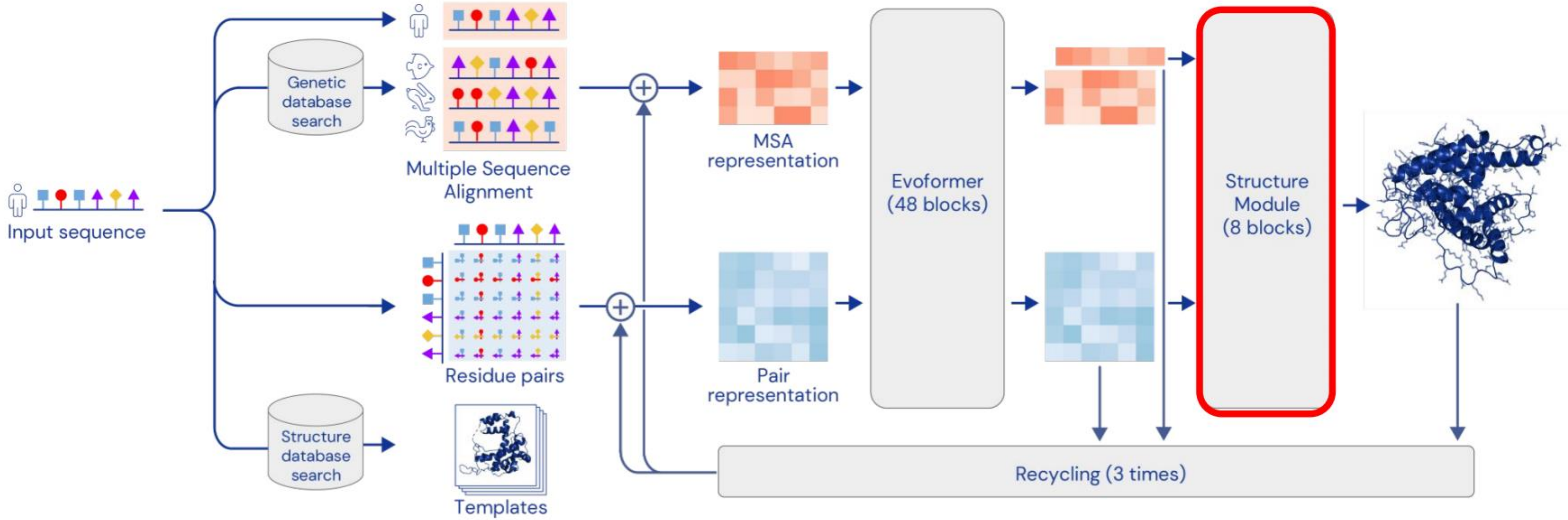


Fig. 3 of Jumper et al. Nature 2021 (the AF2 paper)

Structure Module

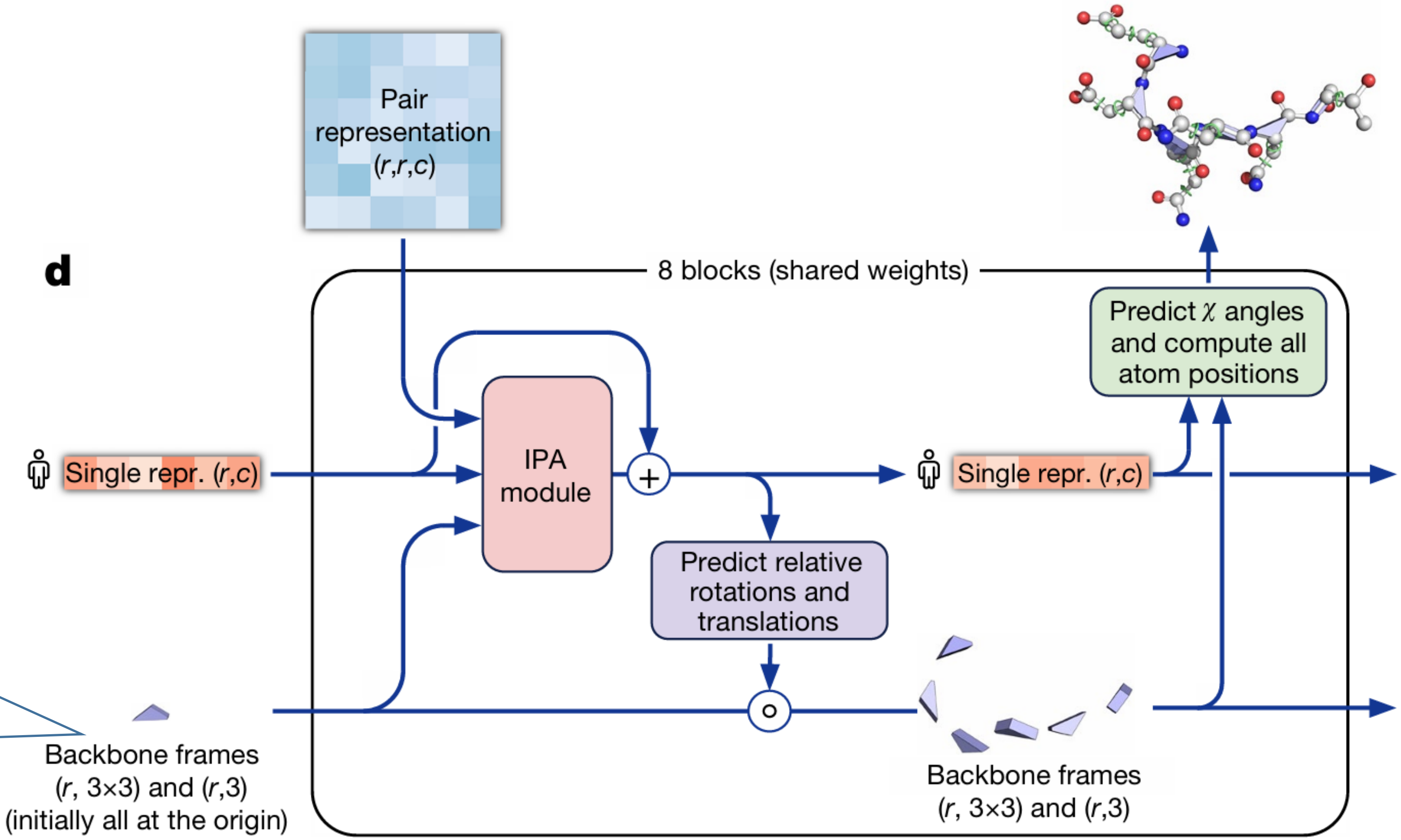
Network



Input sequence
Genetic database search
Multiple Sequence Alignment
Residue pairs
Structure database search
Templates
MSA representation
Pair representation
Evoformer (48 blocks)
Structure Module (8 blocks)
Recycling (3 times)

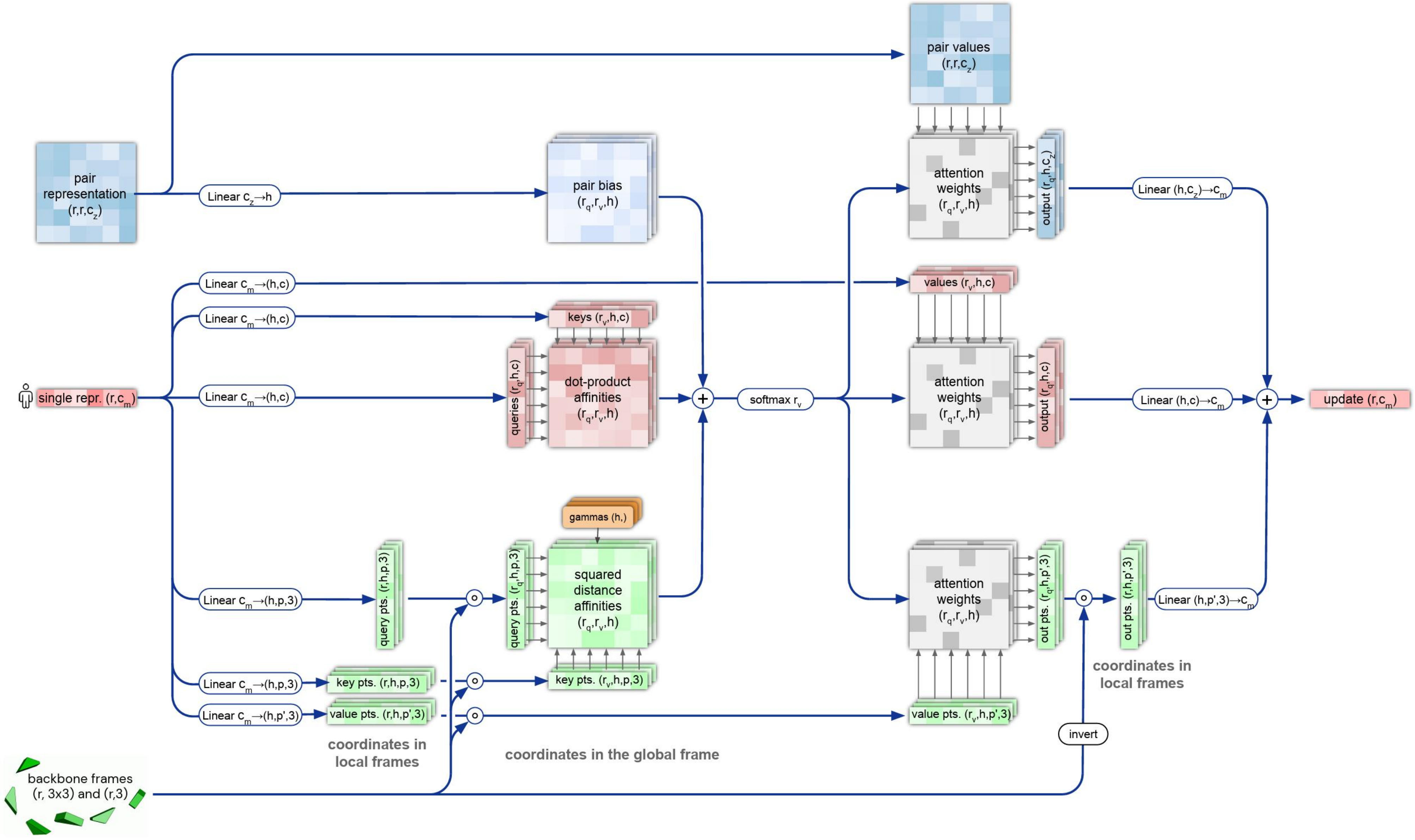
Structure module

The Structure Module maps the **abstract** representation of the protein structure (created by the Evoformer stack) to **concrete** 3D atom coordinates



The "black hole initialization"

Invariant Point Attention (IPA) Module



Structure module

- **End-to-end folding** instead of gradient descent
- Protein backbone = gas of 3-D rigid bodies (chain is learned!)

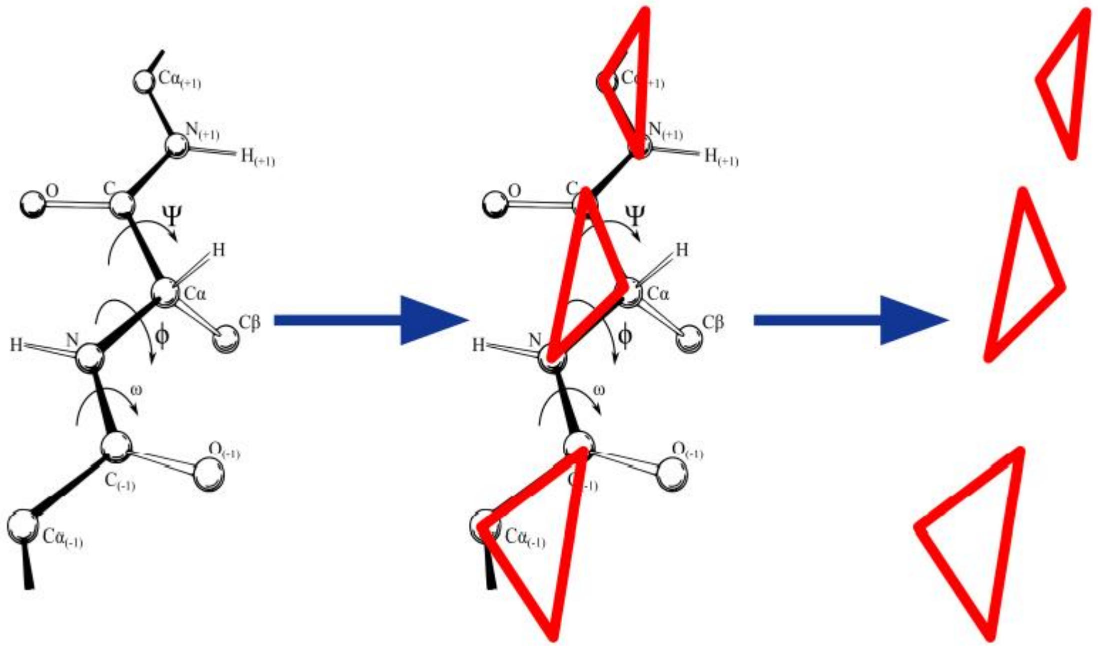
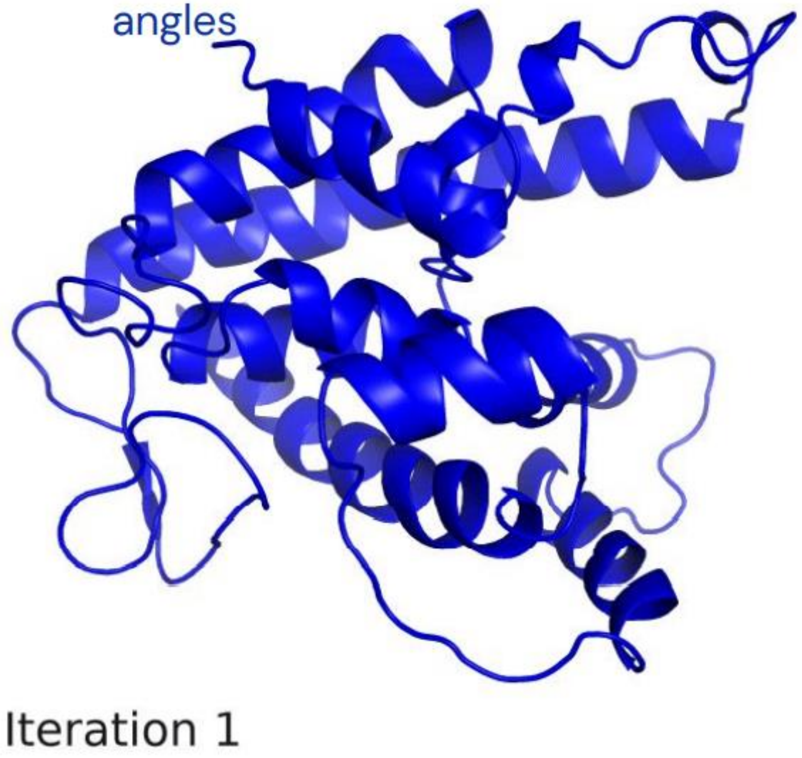


Image: Dcrjsr, vectorised Adam Rędzikowski (CC BY 3.0, Wikipedia)

- **3-D equivariant transformer architecture** updates the rigid bodies / backbone (Invariant Point Attention)
  - Also builds the side chains from torsion angles



Target: T1041

Local frames

- Protein backbone = gas of 3-D rigid bodies (chain is learned!)
- Rigid body orientation defines local coordinate frame along chain

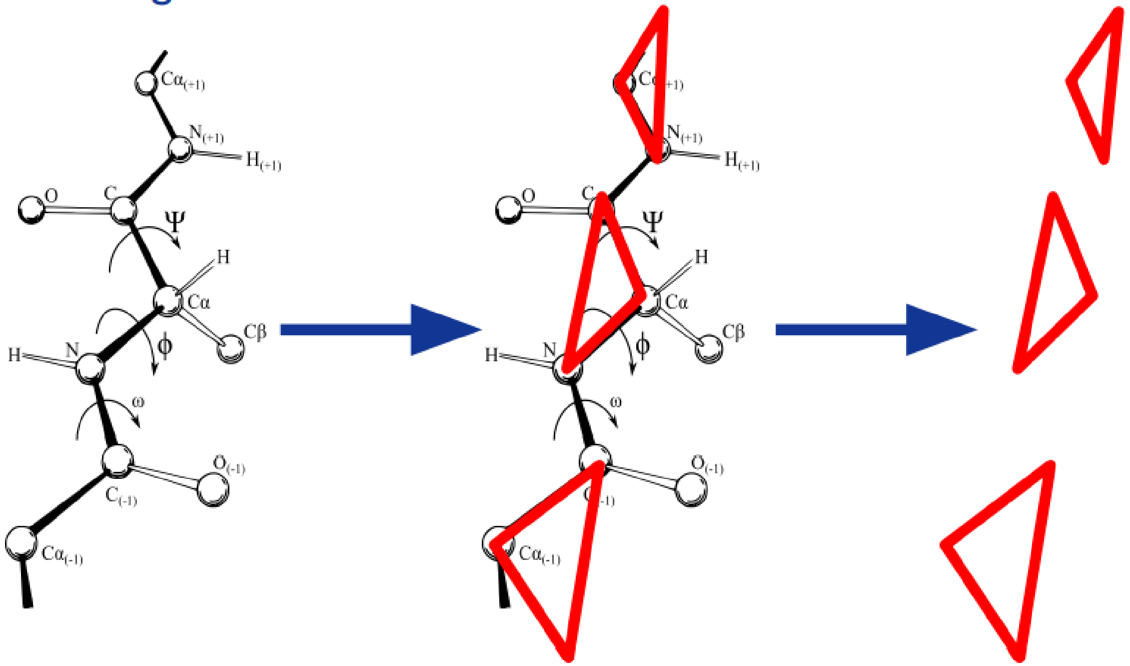


Image: Dcrjsr, vectorised Adam Rędzikowski (CC BY 3.0, Wikipedia)

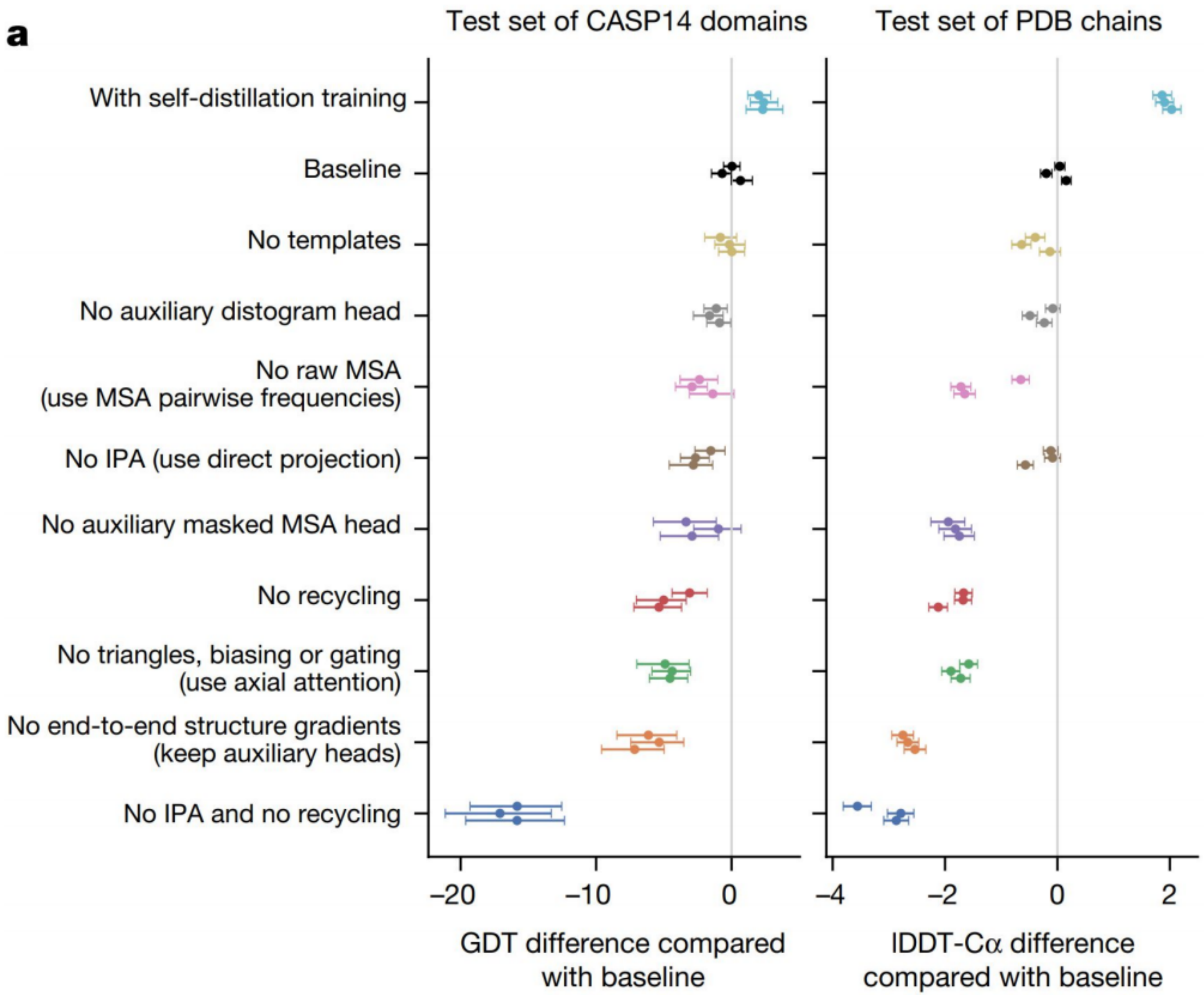
- Coordinates measured in local Frame are **invariant to choice of global frame**

All parts mattered

No single improvement is dominant

More important was the methodology of building the protein intuition into the model

Multiple ablations suggest strong interactions between many of the components

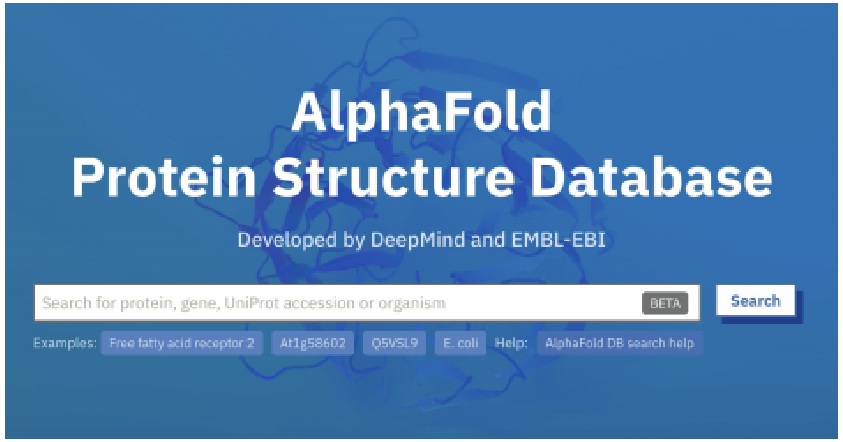


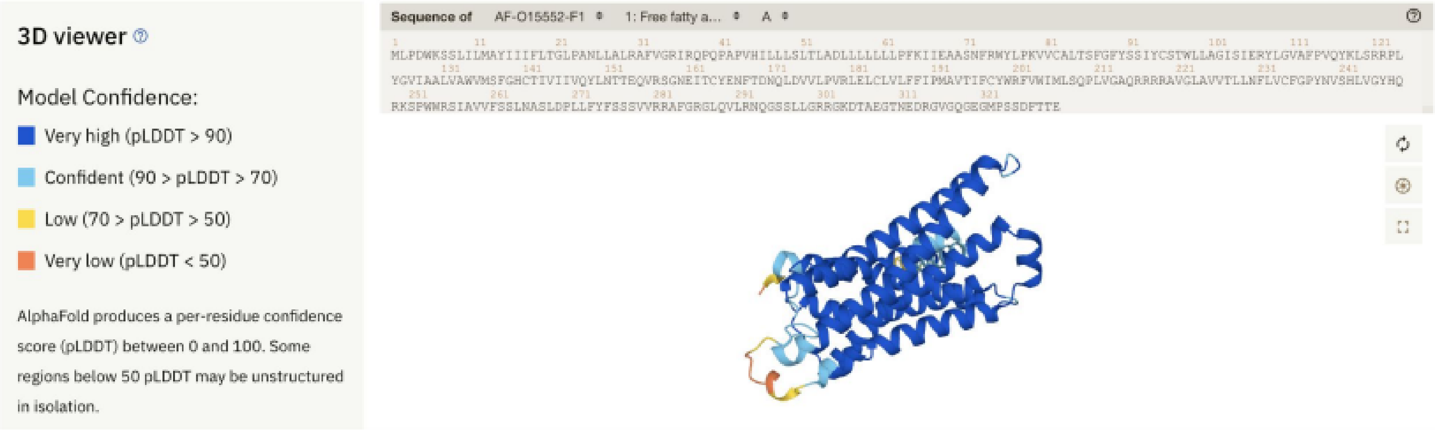
Follow-up Works

AlphaFold Protein Structure Database

Website developed and hosted by EMBL-EBI: alphafold.ebi.ac.uk

Contains pre-run predictions for **21 model organisms**, with plans to expand to **Uniref90** (~100M structures)

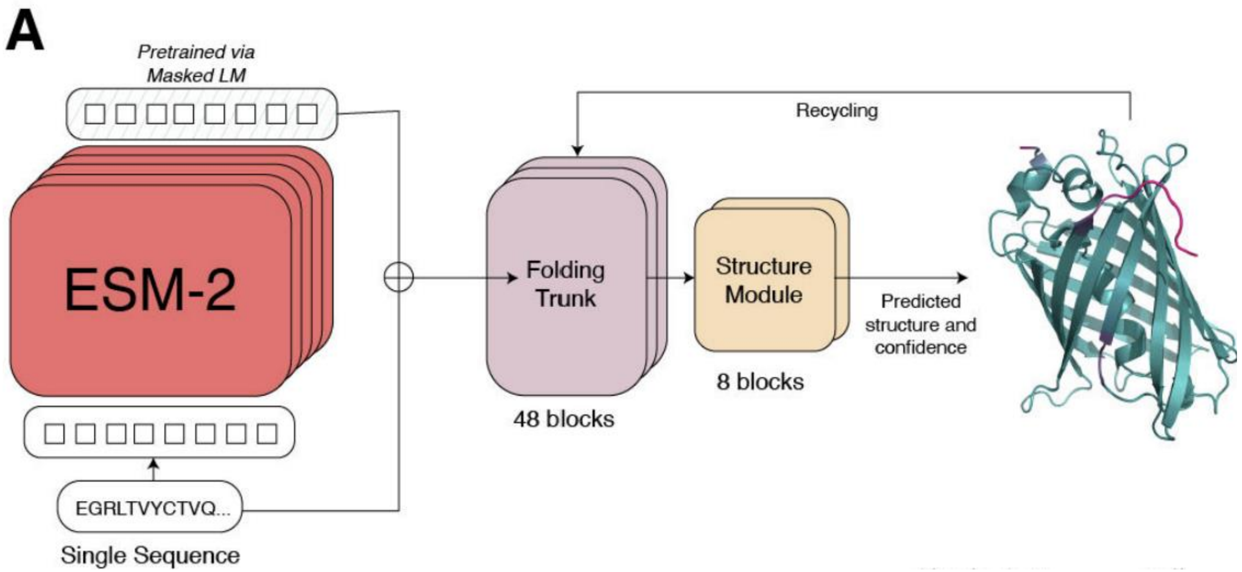


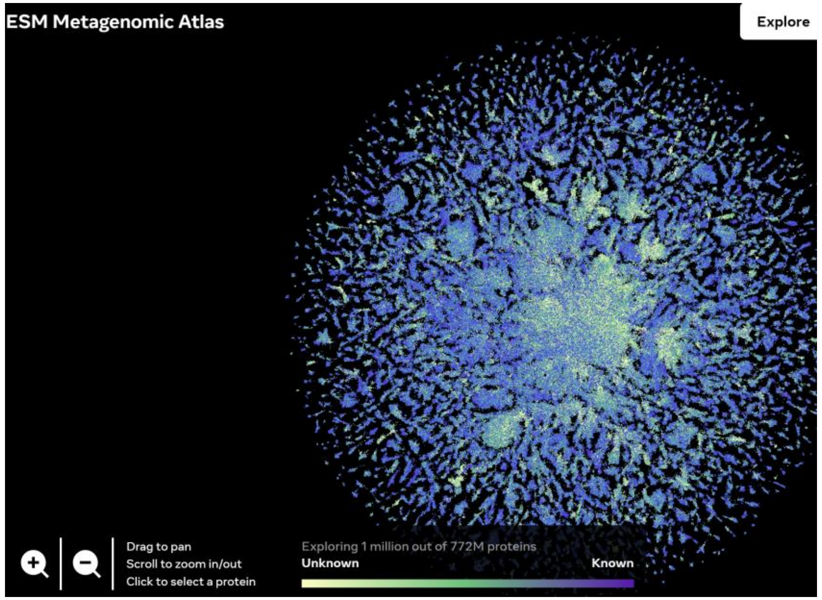


EMBL DeepMind

Protein language models (pLMs) as competitors of AF2

- **AF2**巧妙地将**MSA**作为部分输入特征，并通过**Evoformer**来挖掘同源蛋白序列的共进化信号，但是缺点是搜同源序列需要较长时间，且基于MSA的模型对于序列突变不敏感
- 基于**单序列 pLM**的结构预测模型异军突起，包括OmegaFold，HelixFold-single，ESMFold

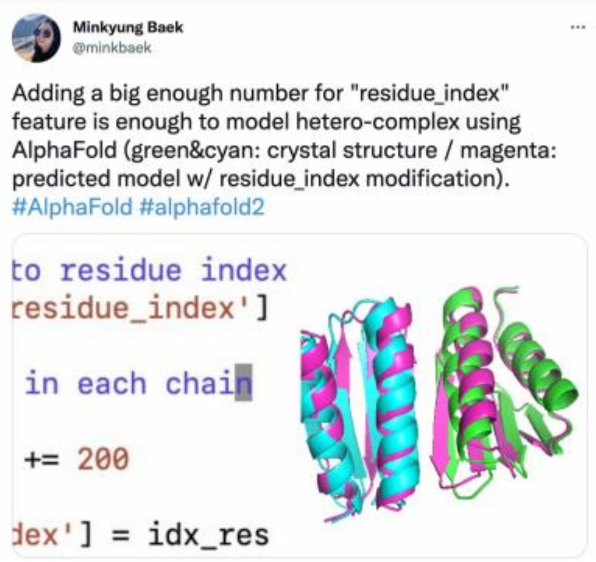


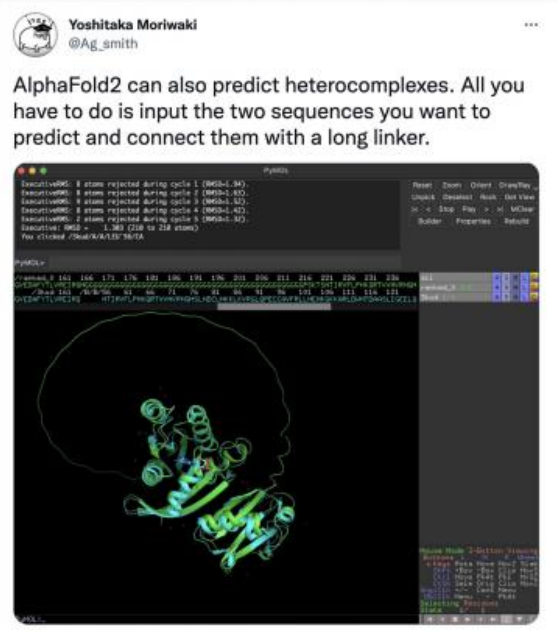


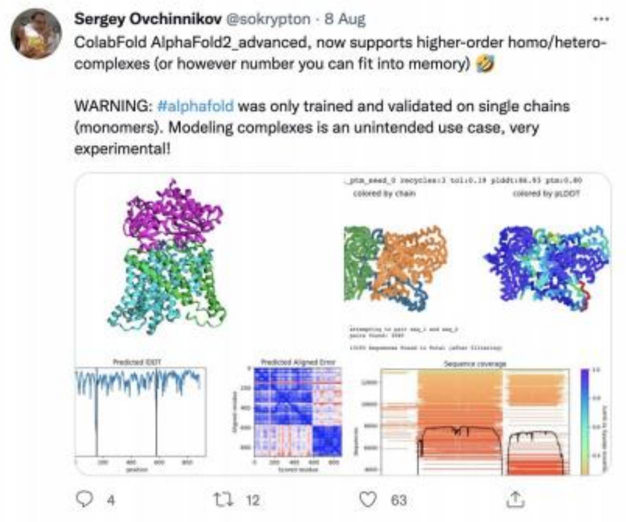
- Meta的 ESMFold，用ESM-2代替MSA，其余结构与AF2一致
- ESMFold的下游任务充分挖掘了预训练蛋白语言模型ESM-2的表达能力

基于ESMFold构建虚拟蛋白数据库ESM Metagenomic Atlas用于辅助药物研发
https://esmatlas.com/

Hacking AlphaFold2 for multimers





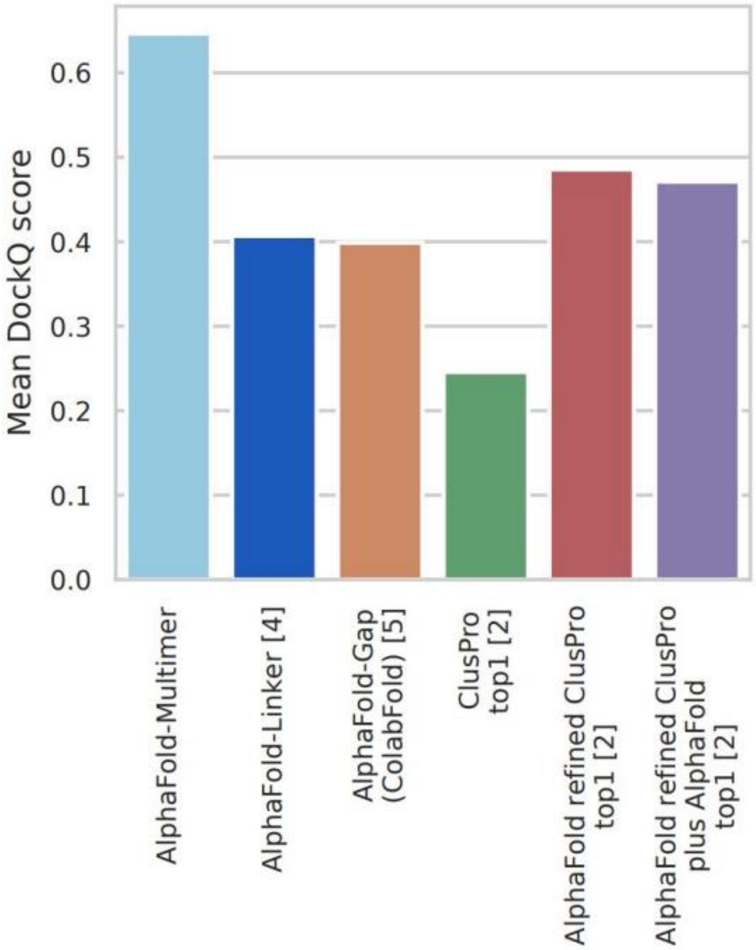


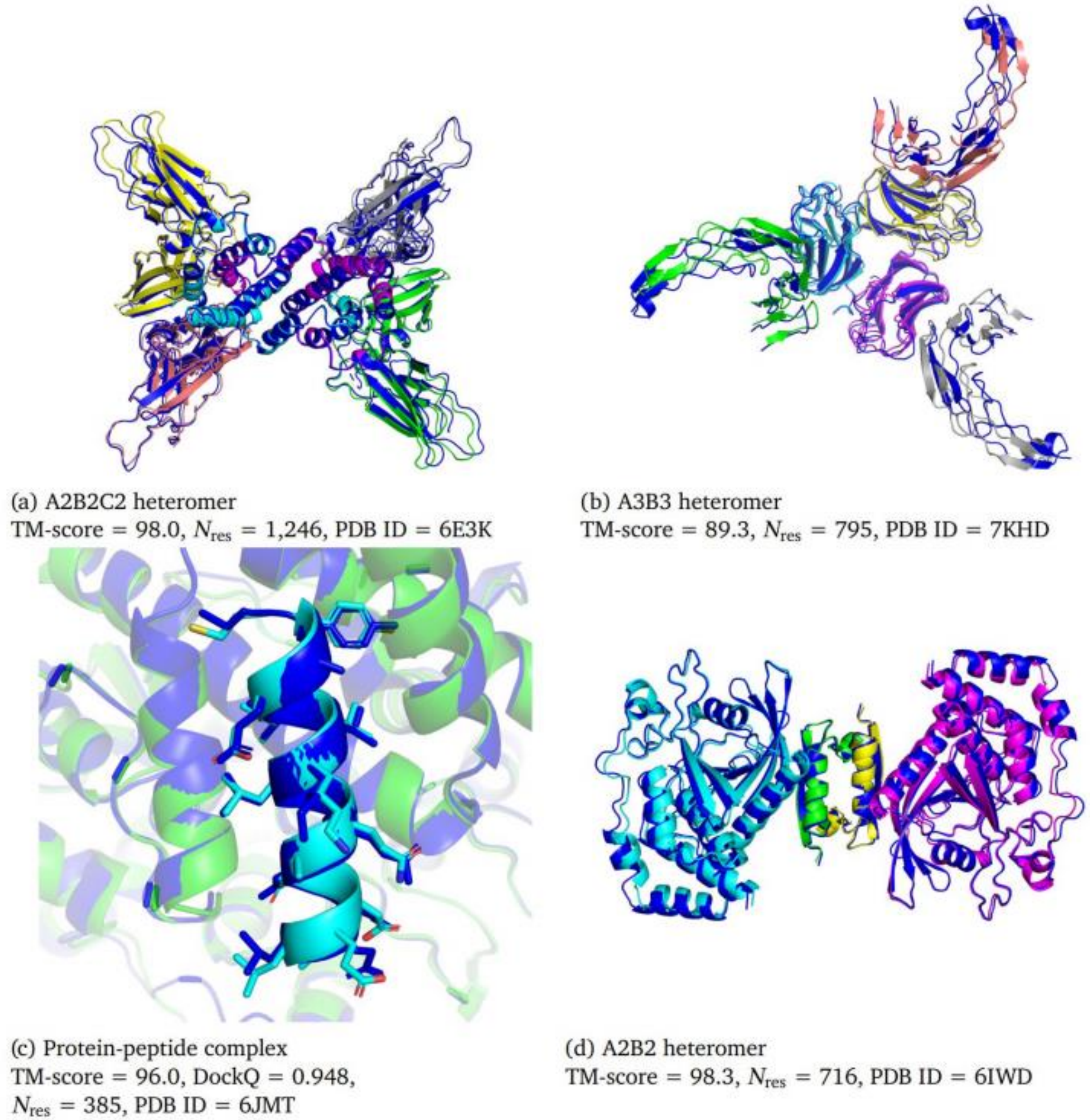
"Even using the default settings, it is clear that **AF2 is superior to all other docking methods**, including other Fold and Dock methods, methods based on shape complementarity and template-based docking"

Bryant, Pozzati, and Elofsson. Improved prediction of protein-protein interactions using AlphaFold2 and extended multiple-sequence alignments, biorxiv 2021. Emphasis ours.

Training AlphaFold to predict multimers (AlphaFold-Multimer)

Adapting the inputs, loss function, and training of AlphaFold to handle multimers and then training the model from scratch





(a) A2B2C2 heteromer
TM-score = 98.0, N_{res} = 1,246, PDB ID = 6E3K

(b) A3B3 heteromer
TM-score = 89.3, N_{res} = 795, PDB ID = 7KHD

(c) Protein-peptide complex
TM-score = 96.0, DockQ = 0.948, N_{res} = 385, PDB ID = 6JMT

(d) A2B2 heteromer
TM-score = 98.3, N_{res} = 716, PDB ID = 6IWD

立志成才报国裕民

Major challenges

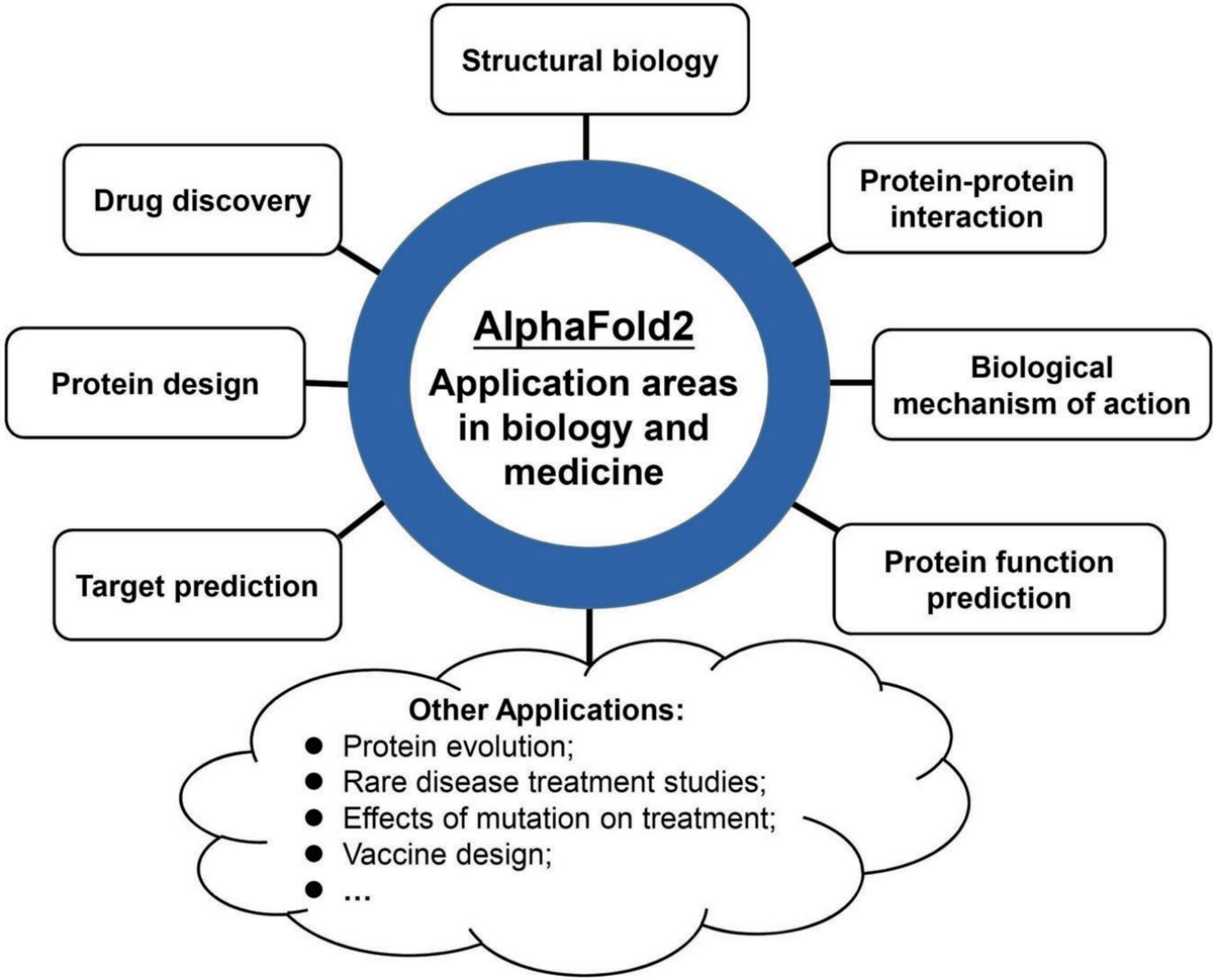
- **Extend AlphaFold2 to other major areas of structure prediction**
  - Increased multimer accuracy and completing the human structural interaction map
  - Structure of nucleic acid and ligand interactions
  - Conformational diversity
- **Structural metaproteome**
  - Extend AlphaFold database to cover UniRef
  - Develop new tools to make these data useful
- **Prediction of mutational effects**
  - Changes in structure and stability
  - Binding affinity

References

- Senior et al. “Improved protein structure prediction using potentials from deep learning”, Nature, Vol 577, pp. 706-710, 2020 (**AlphaFold1 paper**).
- Jumper et al. “Highly accurate protein structure prediction with AlphaFold”, Nature, Vol 596, pp. 583-589, 2021 (**AlphaFold2 paper**).
- Tunyasuvunakool et al. “Highly accurate protein structure prediction for the human proteome”, Nature, Vol 596, pp. 590-596, 2021.
- DeepMind Blog on the breakthrough of AlphaFold2 on CASP14: https://www.deepmind.com/blog/alphafold-a-solution-to-a-50-year-old-grand-challenge-in-biology

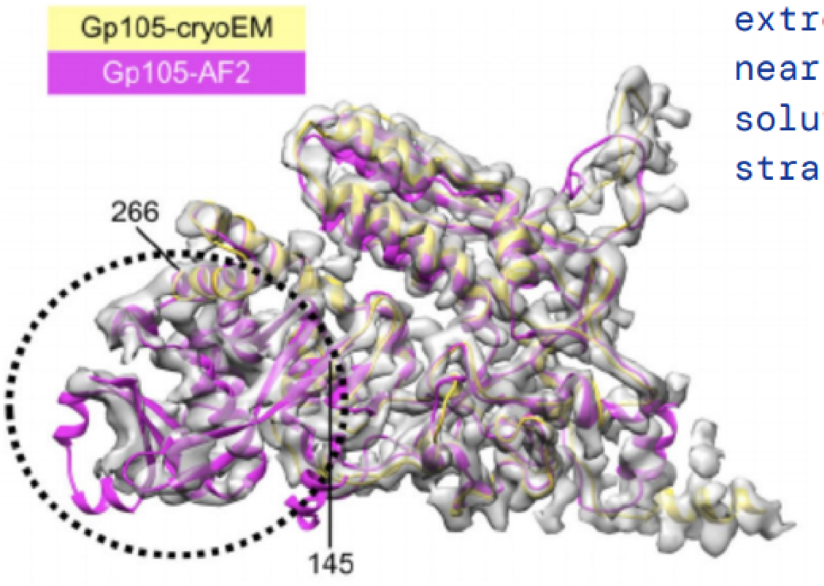
Appendix: Applications of AF2 and protein design

Application of AF2

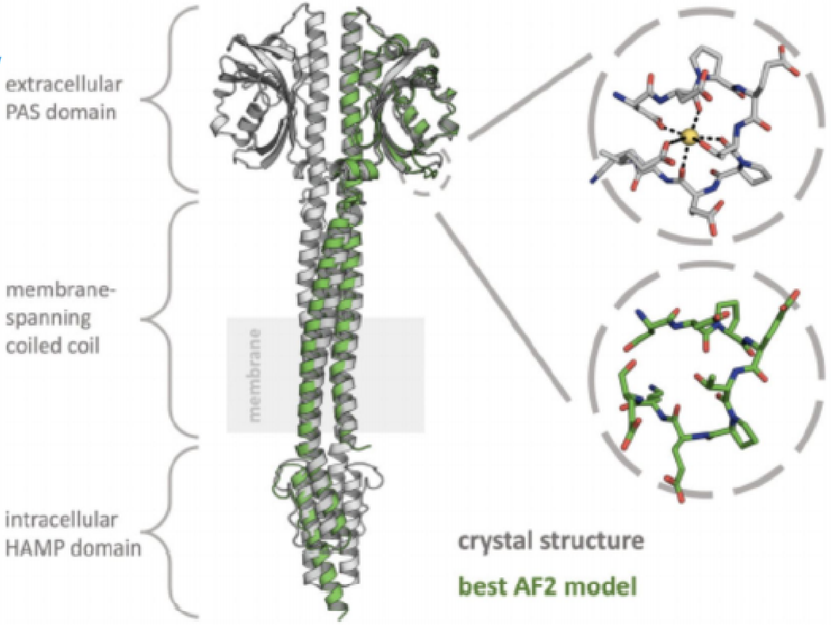


AlphaFold as an aid to experimental structure determination

"Thus, in this case the availability of high-quality structure predictions has reduced an extremely challenging, near-intractable structure solution into a straightforward task."



AlphaFold models of AR9 nvRNAP proteins fit the cryo-EM density nearly perfectly



Crystal structure of dimeric Af1503

Figures taken from: Kryshtafovych, Andriy, et al. Computational models in the service of X-ray and cryo-EM structure determination. Proteins: Structure, Function, and Bioinformatics (2021).

AF2 for drug discovery

上海科技大学 ShanghaiTech University

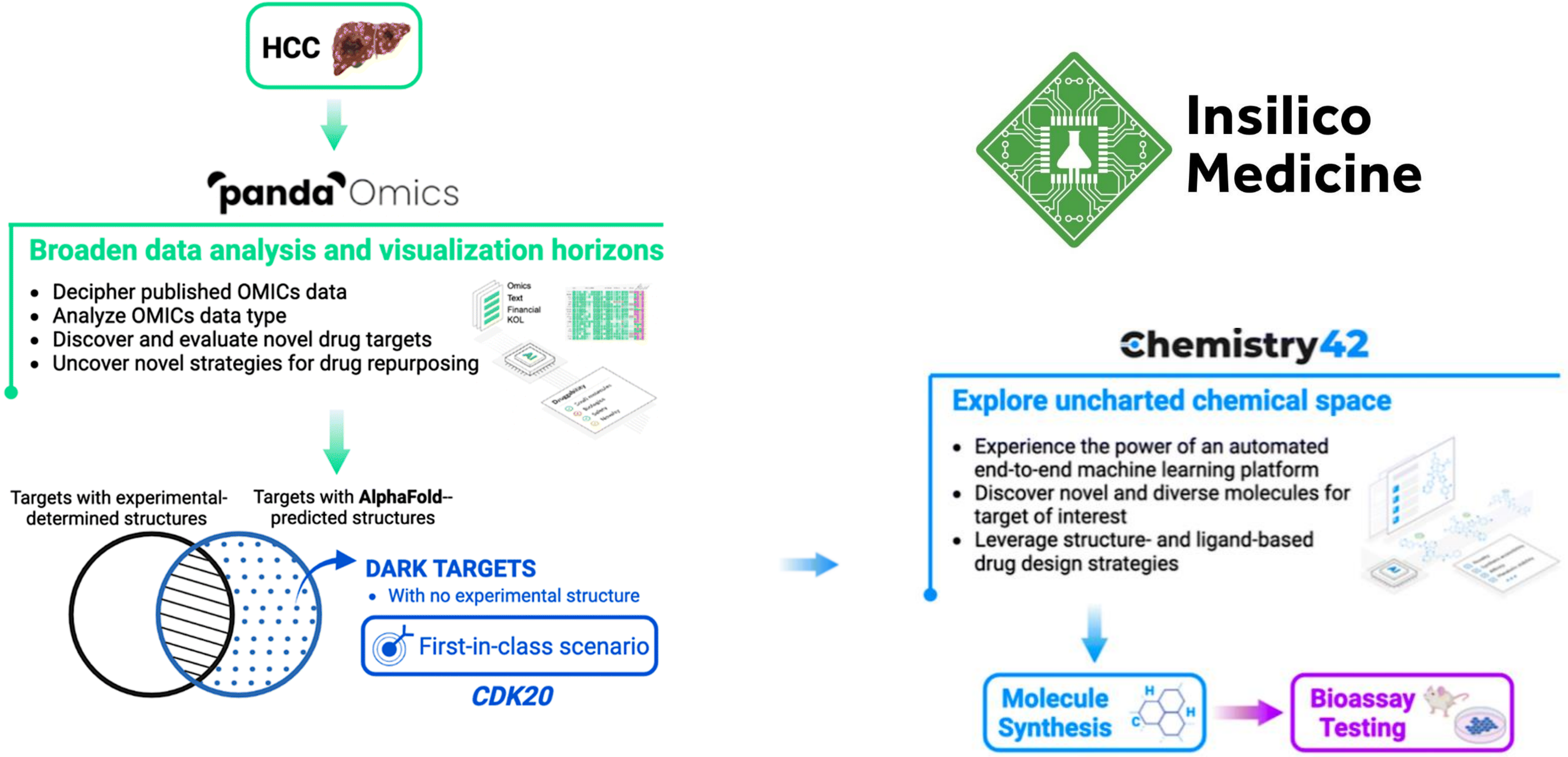
- The study by Zhang et al. demonstrated that **AlphaFold2** (AF2) structures exhibited a similar **enrichment factor** to experimental structures, especially when **flexible docking** was used. These findings indicate that AF2 structures can effectively **replace experimental structures** in **virtual screening**
- Weng et al. applied **AF2** to predict the 3D structure of **WSB1**, a potential anticancer target without available structural information. They optimized the predicted structure through **molecular dynamics** simulations. Then the optimized 3D structure of WSB1 was used as the receptor for molecular docking to screen for WSB1 inhibitors
- Liang et al. identified **JMJD8** as an oncogene associated with **immunosuppression** and **DNA repair**. They used **AF2** to predict the 3D structure of JMJD8 and performed virtual screening to retrieve JMJD8 inhibitors
- Liu et al. proposed a **multi-target drug discovery** method and applied it to therapeutic hypothermia. They used **AF2** and **RoseTTAFold** to predict the structure of related protein targets. Molecular docking was then used to assess protein-drug interactions and determine optimal single drugs or drug combinations

Zhang, Y. et al. Benchmarking Refined and Unrefined AlphaFold2 Structures for Hit Discovery (2022)
Weng, Y. et al. Identification of potential WSB1 inhibitors by AlphaFold modeling, virtual screening, and molecular dynamics simulation studies (2022)
Liang, X. et al. JMJD8 is an M2 macrophage biomarker, and it associates with DNA damage repair to facilitate stemness maintenance, chemoresistance, and immunosuppression in Pan-cancer (2022)
Liu, F. et al. A chronotherapeutics-applicable multi-target therapeutics based on AI: Example of therapeutic hypothermia (2022)

立志 成才 报国 裕民

AF2 in end-to-end Drug Discovery

上海科技大学
ShanghaiTech University



Ren, F. et al. AlphaFold accelerates artificial intelligence powered drug discovery: efficient discovery of a novel CDK20 small molecule inhibitor. (2023).

立志 成才 报国 裕民

Other applications of AF2

上海科技大学 ShanghaiTech University

1. Rare Disease Treatment:
Sebastiano et al. utilized AF2-predicted protein structures to assist in rare disease treatment studies. Specifically, they focused on Alsin, a protein associated with rare motor neuron diseases.

2. Effects of Mutation and Vaccine Design:
Yang et al. applied AF2 to predict the structures of S, N, and M proteins of the Omicron variant of SARS-CoV-2. They analyzed the impact of mutations on specific parts of the S protein (S1 RBD and NTD) and their implications for current vaccines and treatments.

3. Vaccine Design:
Zeng et al. employed AF2 in the design of a hemagglutinin stem vaccine called "B60-Stem-8070." This vaccine demonstrated improved performance compared to the original hemagglutinin stem antigen.

Zeng, D. et al. A hemagglutinin stem vaccine designed rationally by AlphaFold2 confers broad protection against influenza B infection (2022).
Yang, Q. et al. Structural analysis of the SARS-CoV-2 Omicron variant proteins (2021).
Sebastiano, M. R. et al. AI-based protein structure databases have the potential to accelerate rare diseases research: AlphaFoldDB and the case of IAHSP/Alsin (2022).

立志成才报国裕民